Strategic Use of Social Media by Political Parties: Evidence from Italy*

Giacomo Battiston† Federico Boffa‡ Eugenio Levi§ Steven Stillman¶

January 6, 2025

Preliminary draft. Please do not circulate without permission.

Abstract

We examine how Italian politicians strategically use news events and political communication to increase their support in the electorate. Using machine-learning techniques, we extract events about immigration and the economy from newswires. We then test a causal chain connecting events, political communication, the salience of particular cleavages, and support for parties measured. In addition, we measure the relevance of such effects in explaining support for parties by calibrating a structural model of political competition informed by the relationships uncovered by our reduced-form analysis. We show that social media communication about immigration impacts salience and voting, and is more important than actual events in explaining political support.

*We thank participants to UIB, AASLE, SKILS, Bocconi University, Università La Sapienza, ISPP's 2022 Annual Meeting, 6th UniBZ Workshop on Political Economy, 1st Unicatt Political Economy Workshop, CReAM-RFBerlin Workshop 2023, 13th Annual Conference on Immigration in OECD countries, 2023, SISP Conference, 2022, for valuable comments and feedback. We also thank IPSOS and ANSA for providing us data that we employed in the analysis. The results presented in this study are the work of the authors, not ANSA, IPSOS or individual data suppliers. Eugenio Levi's position was funded from 2020 to 2022 by Marie Curie Fellowship CZ.02.2.69/0.0/0.0/19_074/0012727 under the program OP VVV Mobility MSCA-IF of the Ministry of Education, Youth and Sport of Czech Republic.

†Rockwool Foundation Berlin. Email: gb@rfberlin.com

‡Free University of Bolzano-Bozen. Email: federico.boffa@unibz.it

§Masaryk University, Free University of Bolzano-Bozen. Email: eugenio.levi@unibz.it

¶Free University of Bolzano-Bozen. Email: steven.stillman@unibz.it

1 Introduction

Political parties compete in elections where many aspects are uncertain. In particular, voters' priorities shift over time in response to events occurring in society. For example, the recent surge in migration to Europe has benefited far-right populist parties (Hangartner et al., 2019; Dustmann et al., 2019; Steinmayr, 2021). Although such events typically lie beyond politicians' direct control, they can strategically leverage communication to affect the relative salience of different political issues. By timing this communication to emphasize newsworthy events, they can potentially sway public attention towards particular topics that they "own". In the social media era, the connection between political communication and events is likely even more relevant as it is easier for parties to choose when and how to communicate with voters.

This paper examines the traditional and online communication strategies used by the main Italian political parties and their impact in the lead up to and just after the 2018 national election. First, we compare communication by a populist anti-immigration party (Lega) to that by the other main parties at that time. Second, using both reduced-form identification strategies and a structural model of political communication, we examine the link between events, communication, topic salience, and the growth of right-wing populism in Italy during this time period. Our goal is to answer the question of how important the strategic use of social media has been in explaining this populist shift.

While there is a developing literature on the impact of social media communication on polarization and populism, it is not yet clear how political parties make strategic use of social media to exploit events related to topics that they "own". In a multi-dimensional political competition, parties compete not only over voter preferences, but also on the relative weights given by voters to different topics. The emergence of social media has given political parties a new tool that can potential impact salience by shifting attention or by emphasizing social identity (Bordalo et al., 2012; Bonomi et al., 2021). Our paper is the first that examines the full causal chain between events, political communication, topic salience and voting behavior.

We begin by using the universe of political communication by Italian political parties on Facebook, Twitter and in traditional media (excluding TV), along with newswires by the main Italian news agency, ANSA, to measure how party communication responds to different types of events. We classify events and communication using a supervised machine learning approach. Then, using an event history identification strategy, we show that immigration events have larger impacts on the daily communication of the populist anti-immigration party in both traditional and online media, while other parties are more responsive to economic events.

We then use a high-frequency (hourly) event history approach to examine the impact of both traditional and social media communication on the salience of different topics, as measured by

Google search data. In this analysis, we control for the direct impact of different types of events. We find that there is an immediate impact of parties' social media communication, in particular by Lega, on the salience of immigration versus the economy. On the other hand, the use of traditional media has no impact on the relative salience of these topics.

We next examine whether online communication has a causal impact on political support measured by voting intentions. We do this by using a difference-in-RDD strategy that combines time variation in social media communication with geographical variation in 4G coverage.[1] This approach assumes that unobservable time-varying characteristics correlated with political outcomes do not change discontinuously with 4G coverage, for which we provide supporting evidence. We find that Lega's online communication on immigration not only increases the salience of this topic but also attracts voters.

Finally, we quantify the impact of migration events on support for Lega and contrast it with the portion of Lega's support driven by the party's ability to *react* to migration events using online political communication. We do this by specifying and calibrating a structural model of online political communication where two party coalitions compete to maximize their support in the electorate and news events and salience shocks interact with communication to determine the equilibrium salience of immigration versus economic factors. We find that political communication about immigration, rather than immigration events per se, are a more important driver of support for the anti-immigration coalition. In the period leading up to the election of 2018, communication explains about 6.9% of the support for the coalition favoring anti-immigration policies, while events only explain 5.6% of the support.

Our paper is first to examine the impact of communication by political parties on both topic salience and voter intentions. We also expand on the current literature by examining the use of both traditional and online social media and showing that, controlling for both the impacts of events and traditional media, social media communication matters for voting. We contribute to three related areas of the literature.

First, we contribute to the literature examining the causal impact of social media on political outcomes, including polarization (Levy, 2021), voting (Fujiwara et al., 2021), protests (Enikolopov et al., 2020), fake news (Allcott and Gentzkow, 2017), hate crimes (Müller and Schwarz, 2023), and politicians' behavior (Campante et al., 2018).[2] Our paper is the first to establish a causal link between political communication, salience and support for far-right populism. We also contribute to this literature by using a structural econometric model to quantify the overall impact

[1] Informa UK (2013, 2014) and OpenSignal (2017) that show mobile data consumption more than doubled and that download speed was between 3 and 6 times higher under 4G with respect to 3G, and this leads to increased usage of social media.

[2] This literature expands on past work in political science measuring the effects of political communication on TV and in political debates (Bidwell et al., 2020).

of social media.[3]

Second, we contribute to the literature on issue selection and salience by measuring frequent communication choices by parties over a long period of time. Previous papers have mainly looked at political communication in TV debates (Le Pennec and Pons, 2023) and in advertising spaces (Spenkuch and Toniatti, 2018), or have focused on electoral campaigns (Galasso et al., 2023). These papers broadly show that parties persuasion can shift the salience of topics in the final months of the campaign shifts. We find that parties continuously try to affect the salience of cleavages in the electorate, and that, in the case of the populist far-right, this has a direct impact on day-to-day support. These results suggest that social media has extend the persuasion effort by political parties beyond electoral campaigns, debates, and advertising, turning political communication into a continuous campaign.

Third, we contribute to the literature on the effects of immigration on the support for populist parties by accounting for the role of online political communication. Many recent studies have argued that increased immigration explains the recent rise in political populism (Hangartner et al., 2019; Dustmann et al., 2019; Steinmayr, 2021). Others have noted that communication by populist parties was a key ingredient in conjunction with the increased relevance of immigration in explaining the increase support for populism (Bonomi et al., 2021). We contribute to this debate by quantifying the relative relevance of immigration events and communication about immigration in explaining the support for populism, and showing that both make an important contribution.

2 The Italian Political Landscape

Italian political history shares several similarities with that of several other Western countries after World War II. Before 2013, Italy can be characterized by a steady rivalry along ideological lines. From 1947 to 1993, the centrist bloc, led by the Christian Democrats, consistently remained in power. Between 1991 and 1994, both the Communist and the Christian Democratic party dissolved, originating respectively the post-communist Democratic Party of the Left (PDS), and two centrist parties, the slightly more left-oriented Italian People's party (PPI), and the slightly more right-oriented Christian Democratic Center (CCD).

From 1994 to 2013, two coalitions alternated in government. The left-leaning coalition was led at first by the Democratic Party of the Left and then, since its foundation, by the Democratic party (PD), created in 1997 by the merger of the PDS and the PPI. The PD's platform, while exhibiting some variation during the different leaderships, has broadly been in line with main-

[3]Related work by Kendall et al. (2015) use a structural model to examine the importance of information in a particular political campaign.

stream European social democratic parties. On the other side of the spectrum was a center-right coalition led by the conservative Silvio Berlusconi's Forza Italia , which included the CCD and then its heirs.

Our analysis spans the 2015-2019 time period, which marks important changes in Italian politics, most notably the emergence of a populist government. Founded in 1989 as a regionalist northern Italian party with an agenda of decentralization, the Lega gradually turned first into a mainstream conservative party, and subsequently, under Matteo Salvini's leadership (which started in 2013), into a nationalist right-wing party, with a clear anti-immigration platform and a conservative economic ideology and a broader voter base (Vassallo and Valbruzzi, 2018). Additionally, in 2013, the populist Five-Star Movement (M5S), founded in 2005 by Beppe Grillo and characterized by strong anti-establishment traits and a clear anti-politics attitude, started contesting national elections, moving Italy to a de-facto tri-polar political system.

During this time period, there were two general elections. The 2013 general election witnessed the victory, by a small margin, of the center-left coalition. However, the center-left, fell short of a majority of seats in the Senate. After failed attempts to form a government with M5S, which came in third with approximately one-fourth of the votes, a grand coalition involving the center-right (with Forza Italia as the most voted party), center-left and a centrist bloc agreed to form a government, with Enrico Letta from the PD as Prime Minister. In 2014, Matteo Renzi, new leader of the PD, replaced Letta, while part of the center-right abandoned the governing coalition. Renzi then resigned in 2016 and was replaced by Paolo Gentiloni, also of the PD.

In the 2018 elections, the center-right bloc obtained more votes (approximately 37%) than M5S (around 32%) and the center-left (close to 23%), but again not enough to secure a majority of seats. After the slowest government-formation process in the Italian history, M5S and the Lega (without its traditional center-right allies Forza Italia) form a populist coalition government. Tensions between the coalition partners increased during the electoral campaign for the 2019 European elections, and ended with the government collapsing in August 2019, which marks the end of the period of our analysis.

In the years covered by our sample, PD's policy position was rather centrist on economic issues, it had a "Blairian-type" pro-trade and pro-globalization approach along with a moderate view on the amount of taxes and expenditures (Pasquino and Valbruzzi, 2017; Ignazi et al., 2022). It also had a mildly pro-migration policy. Forza Italia, in the opposition throughout the entire period, kept its moderately conservative platform broadly unaltered. M5S both in opposition and governing had a platform which remained that of an anti-elite and anti-caste populist party (Tronconi, 2020). Similarly, Lega's anti-immigration economically conservative platform did not significantly change during the period.

Italy is a particularly good case study for our analysis since the 2018 electoral campaign focused on two main issues, immigration and economic (redistribution and taxing and spending), with both issues seen as being particularly crucial in determining voting patterns (Giannetti et al., 2017). In addition, Lega's platform, being extremely focused on immigration, allows us to place a special focus on that issue.

3 Data

Our analysis covers the period from January 1, 2015 to August 20, 2019. We start in 2015 because of data availability, while the end time coincides with the resignation of the Conte I government, the last one before the outbreak of the COVID-19 emergency, which arguably affected, among other things, parties' communication strategies.[4] Below we discuss the sources of our data on social media messages, news events, salience and polling.

Social media messages. We collected the social media messages by Lega, M5S, FI, PD and their leaders on Twitter and Facebook[5]. In particular, we collected all tweets for "@LegaSalvini" (58,091 tweets), "@matteosalvinimi" (17,334), "@pdnetwork" (650), "@matteorenzi" (1,595), "@maumartina" (550), "@nzingaretti" (563), "@forza_italia" (90,149), "@berlusconi" (3,208), "@Mov5Stelle" (4,662) and "@beppe_grillo" (38,105). We also collected Facebook posts from public pages "Lega Salvini premier" (34,644 posts), "Salvini official" (16,159), "Partito Democratico" (8,188), "Matteo Renzi ufficiale" (1,463), "Maurizio Martina" (323), "Nicola Zingaretti" (814), "Forza Italia" (5,656), "Silvio Berlusconi" (1,357), "MoVimento 5 Stelle" (18,529) and "Beppe Grillo" (17,753).[6]

A machine-learning algorithm trained on this dataset is then used to generate a classification into our two reference categories, economic and immigration, and a third residual group (see Appendix A for more details). Our training sample for this purpose included 3,232 Facebook posts and 4,989 tweets. We run the classification separately for tweets and Facebook posts to account for the differences in length and communication style across the two medias.[7] In our

[4]Google trends hourly data are not available until 2015.

[5]PD changed three leaders during the period under consideration. There was Matteo Renzi from the start of the period until the 12th March 2018, then Maurizio Martina until the 17th November 2018, and finally Nicola Zingaretti. For each leader, we only considered their social media and traditional messages in the period in which they were secretary of the party.

[6]Given the download limits on the Twitter public API (3200 tweets per user), we develop an algorithm to collect the universe of tweets. We first identify all tweets IDs by a specific user using the Python *selenium* module, then download each of them one by one. Facebook data comes from Crowdtangle, Facebook's interface for data sharing with commercial and academic ventures.

[7]In further analyses, after having classified based on the topic, we also look at sentiment. We employ a sentiment classifier trained on a corpus of Italian tweets to do our sentiment classification (Bianchi et al., 2021). The outcome of this classification suggests that, across all topics and parties, over 80% of the social media messages

empirical analysis, we aggregate the leaders' and the corresponding party's messages.

Figure 1 in the top panel depicts the- daily total number of social media messages over time for all parties.[8] It shows that Lega, M5S and Forza Italia communicate way more than the PD throughout the period. On average, Lega and Salvini produce 74.7 (32.4 s.d.) social media messages per day between Facebook and Twitter, while the corresponding figures for the other parties are 59.2 (45.3) for Forza Italia, 46 (55.7) for the M5S, and 8.3 (6.5) for PD. Furthermore, Lega exhibits a spike close to three events of major national relevance in Italy (identified by the dashed vertical line in the graph). In general, Lega communicates more during the summer, likely as a result of the higher intensity of immigration events, such as rescues and arrivals episodes. Communication by the M5S follows less regular patterns and is mostly driven by its founder Beppe Grillo's tweeting behavior, which concentrates on scandals involving his political enemies. The communication by Forza Italia broadly mirrors that by Lega, although it shrinks after the 2018 election.

The top panel of Figure 2 reports the evolution of the difference between immigration and economic messages. Lega is the only party that communicates more about immigration, as it produces on average 2.2 more messages per day on immigration than on the economy. The difference is especially large in 2015 during the refugee crisis, as well as during the Conte I government, when its leader Salvini was Minister of Internal Affairs. All other parties concentrate their communication efforts more on the economy throughout the whole period[9]. We interpret this, consistently with our theoretical assumptions, as Lega owning the migration issue, and the other parties fighting for ownership on the economy.[10]

Traditional messages. We derive traditional media messages from the universe of newswires provided by ANSA, the main Italian news agency. ANSA is a politically neutral news agency. We got permission to access ANSA MIDA, a professional portal for the ANSA database containing all newswires from 2015 onwards. We scraped it using a Python algorithm based on the *selenium* module. Our initial dataset included 1,802,687 newswires from 2015 to 2020. We identified messages by the party leaders (Salvini, Renzi, Martina, Zingaretti, Berlusconi and Grillo) by

have a negative sentiment. That is why we do not elaborate more on this issue.

[8]We smooth out the time series using a non linear filter with a median smoother of span 7. The vertical dashed lines represent, in chronological order, the start of the 2015 refugee crisis, the 2016 referendum on constitutional reforms, the start of the 2018 general elections' electoral campaign, the 2018 general election, and the formation of the Conte I government (supported by Lega and the M5S).

[9]M5S has 5.7 more messages per day on the economy than on migration, PD 2.3 and Forza Italia 10.2.

[10]The content of messages broadly reflects the parties' policy positions. In Appendix Table D1 we present the most frequent bigrams for each party by topic. Lega's messages on migration focus on illegal migration, security and "presumed refugees" while PD is more concerned with the activity of the coastal guard and saving human lives. Forza Italia's messages broadly mirror those by Lega while M5S's messages lack a clear focus. On the economy, a left/right divide emerges, with Lega and Forza Italia focusing on fewer taxes and PD and M5S on jobs, minimum income and citizenship's income.

exploiting ANSA's peculiar way of titling the newswires[11]. When the title has the surname of the politician either as the first word or after a colon ending a description of the topic of the newswire, then it contains a message by the leader itself. Most of these messages represent direct communication of politicians that later in the day gets picked up by newspapers or TV news, while a handful of them reports politicians' statements from newspaper or TV's interviews. The machine-learning model we used for social media messages is then used to generate a classification into our two reference categories, economic and immigration, and a third residual group.

The bottom panel of Figure 1 depicts the daily total number of traditional media messages over time for all parties[12]. Forza Italia and M5S communicate very little in this traditional way - on average 0.2 and 0.3 messages per day, respectively[13]. For Lega and PD this communication is more frequent with 1 and 1.4 messages per day respectively. However, traditional messages are less frequent than the social media ones by an average ratio of 77.8 and 6.2 respectively. Interestingly, PD communicated more than Lega while in government, and vice versa, probably explaining the rise in social media communication for PD in more recent times in our sample period. When we look at topics, M5S and Forza Italia only communicate on the economy[14]. As expected, Lega communicates relatively more on migration and PD more on the economy.

Events. We measure the occurrence of politically-relevant events using ANSA. To the extent of our knowledge, our paper is the first to employ newswires to extract events. At this purpose, we dropped newswires unrelated to the economy and immigration (e.g., weather reports, daily recaps, summaries of election results, stock market news, foreign news) exploiting ANSA's own classification. We further excluded politicians' statements using a dictionary-based approach.[15] This left us with 552,899 newswires, which we classified using the same method employed for political messages using supervised machine learning (see Appendix A for details).

We then adapted the clustering algorithm by Piskorski et al. (2008) to extract events from news. Intuitively, we cluster pieces of news based on the similarity of their text, and we consider clusters of news to be generated by the same underlying events. We then take the timestamp on the first occurring article to measure when the event occurred.[16] The bottom panel of Figure 2 shows

[11]We only consider the party leaders and not the official party communication because on traditional media that is very scarce (and noisy, often including communication by local party branches).

[12]The only difference with the top panel of Figure 1 is that we now smooth out the time series using a non linear filter with a median smoother of span 28 rather than 7.

[13]We conjecture that this depends in the case of Forza Italia on Berlusconi's ownership of TVs and in the case of M5S on their populist way of communicating online only.

[14]From our classifier, M5S has 2 traditional messages on immigration over 4 years, Forza Italia not even that.

[15]We create a list of politicians including all national MPs elected in 2008, 2013 and 2018, all European MPs elected in 2014 and 2019, members of Renzi's, Gentiloni's and Conte I's cabinets, all regional Presidents, all majors of provincial capitals in office between 2015 and 2020, all trade unions' and business associations' leaders, all leaders of the Italian Roman Catholic Bishops' Conference, as well as the Pope and the last two Presidents of the Republic.

[16]More precisely, we perform agglomerative (hierarchical) clustering on data with average cosine distance;

the pattern of events over time with economic events always more common than immigration events (77.7 vs. 12.7 per day on average).

Salience. We measure salience of immigration versus the economy using Google search volume in Italy, downloaded through Google Trends APIs. We first build a dictionary of the most representative 10 keywords for each topic. To this end, we consider the top 10 words in terms of *feature importance* that emerged while dealing with the classification of events.[17] The list of words by category is available in Appendix B. We also perform a robustness in which we use 20 keywords rather than 10. Then, we collect hourly searches by computing Google Trends queries involving the keywords using the Python module *pytrends*[18]. We then rescale by a common factor time series representing different words pertaining to the same category and then sum them up.

Polling data and 4G coverage. We use polling data collected by IPSOS, a leading Italian polling company from 2015 to 2018. Each week IPSOS surveys via phone a sample of 1000 respondents, selected through the using CATI statistical methodology. The overall sample aggregates to approximately 50,000 observations per year[19]. Each observation in this dataset is an individual response and includes information on voting intentions and intention to vote, as well as sociodemographic characteristics - age, gender, education, and occupation. Importantly, the data includes the respondent's municipality of residence, which we use to match each respondent with the availability 4G coverage in their area collected via the Collins Bartholomew map explorer (https://www.collinsbartholomew.com/mobile-network-coverage-map-data-technical-detail/).

agglomerative clustering is restricted to links closer than 2 weeks. In order to deal with disconnected components, we separately run clustering for different network components of the connectivity matrix. The algorithm then prepares a loop creating adjacency matrix based on cosine similarity at each time t. We choose a threshold of 0.70 to determine the maximum distance, but in the analysis we perform robustness over several thresholds and the results stay qualitatively the same.

[17] *Feature importance* is a procedure that extracts from a classification model the words that better predict classification into a category.

[18] Given that Google allows for contemporaneous queries of 6 words only, we had to perform multiple overlapping queries where the overlap was both within and between the categories. In addition, Google only returns data on a hourly basis as opposed to daily, weekly or monthly if the time range of the query is equal or lower than a week. Also, Google Trends time series are not comparable over time if there is no overlap between dates. We thus had to instruct our algorithm to overlap weekly queries over time and continuously rescale them over time. The overlap was of 3 days, so to accommodate for values of the series close to the end/beginning of the range equal to zero.

[19] Aggregated data is publicly available at www.sondaggipoliticoelettorali.it

4 Reduced-Form Evidence

We first look at the reduced-form relationships between (i) events and online political messages, (ii) political messages, events and salience, and (iii) political messages, events and intentions to vote. In each case, we use an identification strategy designed to reveal causal relationships.

Events and political messages. We first investigate the impact of events on party's online and traditional communication. More specifically, we look at if and how much political parties communicate on immigration or on the status of the economy just after a migration-related or an economy-related event occurs. In the Appendix, we examine the residual category of messages mostly as a placebo, although messages in this category could well be complements or substitutes for messages on immigration or on the economy.

In particular, we estimate the following specification:

$$messages_{k,t,p} = \delta_0 + \sum_{i=1}^{2} \delta_i messages_{t-i} + \sum_k \delta_k events_{k,t} + \delta_t + \varepsilon_t \tag{1}$$

where t indexes the day. The variable $messages_{k,t,p}$ reports the total standardized number of either social media or traditional messages in a given day t for topic k by party p. Messages by topic and by party are standardized over the total amount of either social media or traditional messages so that all coefficients are directly comparable. The standardized variable $events_{k,t}$ reports the number of events of category k in the same day. We include in each regression two day lags of the total amount of messages $messages_{t-i}$ and a wide range of time fixed effects δ_t, more specifically day of the week, week of the year, and year fixed effects as well as daily time trends that vary by year. Table 2 shows results from this regression analysis. In the top panel, we show results for Lega and M5S, while FI and PD are in the bottom row. Results on social media messages are in the left columns, while those on traditional messages to the right.

We first look at Lega. Column (5) shows that a 1-point standard deviation increase in immigration events leads to a statistically significant 0.027 standard deviation increase in Lega social media messages on immigration. As expected, it has no discernible effect on tweets and post on the economy, as indicated in Column (6). The same effect on traditional media in Column (7) is of 0.016 s.d., so discernibly lower than that of social media. Again, there is no discernible effect on traditional media on the economy, as indicated in Column (8). Economic events show a similar pattern. They lead to more messages on the economy by Lega; a 1-point standard deviation increase in events corresponds to a 0.012 s.d. increase in social media messages. This effect is smaller than that of migration events on migration messages, suggesting that Lega indeed exploits events related to migration for its communication strategy more than those related to the

economy. Again, as before, and intuitively, economic events do not significantly affect messages on migration. More interestingly, they also do not significantly affect traditional messages on the economy, consistently with Lega's populist approach of favouring social media over more institutional channels.

When looking at the other parties, a few results stand out. First, as expected and consistently with the notion that Lega "owns" the immigration issue, the other three main parties react more to economic than to immigration events. Interestingly, M5S reacts to immigration events by increasing its political communication on the economy. While this result may at a first glance appear counter-intuitive, it is consistent with qualitative evidence on M5S communication strategies that show that M5S tries to shift the focus from immigration policies to the immigrants' impact on jobs and economic insecurity (Levi et al., 2019).[20] We also find that PD reacts relatively more on traditional media than on social media, especially on the economy.

Figure D1 and D2 in the Appendix plots event-study graphs on social media and traditional messages using the same specification as in Table 2, but now allowing for 7 days lags and leads in the explanatory variables, i.e. economic and immigration events, to account for potential anticipation effects of a news (e.g., a report due in a certain date), or for events that trigger a long lasting debate extending to days after their occurrence. The picture on social media messages is consistent with results from Table 2, and it further shows that most of the effects are short-lived and concentrated in the same day when the event happen. The same cannot be said of traditional media messages, as the effects become noisier and less concentrated in the same day the events happened. In Appendix D, we provide robustness checks by looking at important events only and at Facebook and Twitter communication separately. When looking at important events only in Table D2, results go in the same direction; however, coefficients are smaller suggesting that our main results are not driven by the important events. We find similar impacts on Facebook and Twitter communication (see Table D3), although parties seem to have a strategic use of each media. We also investigate in Table D4 and Table D5 whether our results are driven by the political parties' strategy in specific time periods, like the electoral campaign, or are related to their incumbency status. Results on Lega's social media messages don't, while we find some evidence that the increase in traditional messages on migration is driven by the period in which Lega was in government. We also find limited evidence that other parties are more reactive while in opposition and during the electoral campaign.

Social media messages, events and salience. We next examine the effects of social media messages on topic salience, measured as the difference between the volume of searches on immigration and on the status of the economy.

[20] We also find that a large fraction of the increase in the messages by M5S on the economy following immigration events is associated to events involving immigrant exploitation in the labor market (see Appendix C).

In particular, we estimate the following event-study specification:

$$salience_t = \delta_0 + \sum_{i=1}^{196} \delta_i salience_{t-i} + \sum_{k,p} \delta_{k_m,p} \sum_{i=0}^{24} messages_{t-i,k,t,p} + \sum_{k} \delta_{k_e} \sum_{i=0}^{24} events_{t-i,k,t} + \delta_t + \varepsilon_t \tag{2}$$

where t indexes the hour. The dependent variable $salience_t$ represents the standardized difference in hourly Google searches between immigration-related and economy-related issues and we control for 196 lags of the dependent variable amounting to a week. The main variable of interest is $\sum_{i=0}^{24} messages_{t-i,k,t,p}$ reporting the total standardized number of messages in the previous 24 hours for topic k by party p at hour t. We include social media and traditional messages separately in our regressions. Again, messages by topic and by party are standardized over the total amount of either social media or traditional messages so that all coefficients can be directly compared. The standardized variable $\sum_{i=0}^{24} events_{t-i,k,t}$ controls for the number of events of category k in the previous 24 hours. Time fixed effects δ_t are in this specification date and hour of the day times day of the week fixed effects, so that our estimates come from comparisons in Google searches within the same day controlling for within-day trends.

Table 3 shows results from this specification in column (3) and (4). In column (1) the volume of messages aggregated by topic $\sum_{i=0}^{24} messages_{t-i,k,t,p}$ is additionally summed across the parties p and we do not control for events. In column (2) we bring back the rolling sum of events in the previous 24 hours, to disentangle the direct impact of events on salience from the impact on salience mediated by the parties communication. It is to note that we only report coefficients for social media messages: all coefficients associated to traditional messages, although pointing in the right direction, are not significant.

Results in columns (1) and (2) broadly suggest that political communication affects salience in the intuitive direction, although differentially for immigration-related and for the economy-related messages. An increase in the volume on messages on immigration increases the salience of immigration. The effect remains strongly statistically significant, and coefficients remain quite stable, even after the inclusion of the events in column (2). In terms of magnitudes, a one-standard deviation increase in communication on immigration increases Google searches on immigration over the economy by 13-17% of its standard deviation. An equivalent increase in messages on the economy, instead, has no discernible effect on relative salience.

In column (3), we unpack social media messages by party. We find that the significant effect on messages on migration on salience is driven by communication by Lega. A one-standard deviation increase in communication increases the relative salience of migration over the economy by 23.89% of its standard deviation. We also find a weakly significant impact of Lega

communication on the economy, with a 17.19% effect in the opposite direction. An effect of PD messages on migration is also weakly significant. In column (4) we unpack events by their type, such as delivered to us by a Latent Dirichlet Allocation (LDA) over the ANSA database with 14 categories for each topic. In Appendix C, we specify the details of the LDA clustering procedure and its outcomes. Here, it is sufficient to say that the inclusion of events by more detailed type provides more precision in the estimates and accounts for the possibility that messages are confounded with specific types of events. Results mostly stay the same. The impact of Lega messages on the economy now becomes statistically significant at 5% level and the effect of PD messages on migration becomes significant at 5% level too.

Table 3 shows that Lega communication is more effective at affecting salience than communication by the other parties. To test this formally, we perform F-tests after regressions in columns (3) and (4) that coefficients associated to the communication of each party are jointly zero regardless of the topic. Indeed, what we find is that we can reject the null hypothesis at 1% level for Lega in both regressions while we cannot do the same for the other parties. Overall, Lega's communications affect salience more than those of other parties. Moreover, its focus on migration seems to pay off by increasing the salience of immigration as compared to the economy. M5S, FI and PD do not seem to achieve the result of shifting salience on their own preferred issue, i.e. the economy.

We now look at the robustness of these results to different specifications of the regression models. We start, as a baseline model, from column (4) of Table 3. We first examine whether our results are robust to using a 12 hour timeframe as opposed to a 24 hour time frame. Next, we consider Google searches time series built with queries over 20 keywords for each category instead of 10 (see section on data and Appendix B for details). Finally, one may worry that events, no matter how accurately we controlled for their type in column (4) of Table 3, may still confound the influence of messages on salience. To test this, we drop all observations where an event occurred in the previous 24 hours. Given the limited number of remaining observations (91), and the consequent risk of saturating the model, we include in this regression only 12 lags of the dependent variable. Results are in Table 4. They are fully consistent with the previous results.

Political messages, events and intentions to vote. The previous analyses suggest that parties leverage events to increase the effectiveness of online political communication and change the salience of particular issues. Using data on polls, we now investigate whether their communication ultimately affects citizens' propensity to vote for a given party. Employing only time variation in political communication and polls is problematic in this case. Since polling data frequency is daily at best, overcoming endogeneity concerns with frequent data is not a viable option. Instead, we used difference-in-RDD design, combining time variation in party posts with an exposure variable based on 4G coverage in the year prior to the treatment, measured

by Collins Bartholomew (https://www.collinsbartholomew.com/mobile-network-coverage-map-data-technical-detail/) . We consider 4G coverage as a proxy of social media usage in line with reports that find that mobile data usage more than doubled under 4G with respect to 3G with specific increases in Facebook usage (Informa UK, 2013, 2014)[21].

More specifically, we match IPSOS survey data on a given day to communication by parties in different topics (economy, immigration, and the residual category), and events in a given category (economy, immigration) all in the prior week. We estimate the impact of communication and events by measuring the differential effect of polls and communication across areas with 4G versus 3G mobile internet coverage, controlling for aggregate shocks in the support to a given party (week FEs), time-invariant characteristics political characteristics (geographical FEs) and potential selection of the sample of respondents (through time-varying averages of respondents' characteristics).

In particular, we estimate the following specification:

$$
\begin{aligned}
y_{i,l,t} = \delta_0 &+ \sum_k \delta_k^e events_{k,t} \times 4G_Coverage_l + \\
&+ \sum_j \sum_p \delta_k^m \#messages_{p,j,t} \times 4G_Coverage_l + \\
&+ \sum_k \delta_k^e events_{k,t} \times 4G_Distance_l + \\
&+ \sum_j \sum_p \delta_k^m \#messages_{p,j,t} \times 4G_Distance_l + \\
&+ \sum_k \delta_k^e events_{k,t} \times 4G_Coverage_l \times 4G_Distance_l + \\
&+ \sum_j \sum_p \delta_k^m \#messages_{p,j,t} \times 4G_Coverage_l \times 4G_Distance_l + \\
&+ \delta_l + \delta_t + X_i + \\
&+ \varepsilon_{i,l,t}
\end{aligned}
\tag{3}
$$

where l, i, and t, index the municipality, individual, and the week, respectively. The variable $events_{k,t}$ reports the number of events of category k in a the prior week. The variable

[21] Furthermore, independent data from Italy in 2017 indicates that download speed under 4G with respect to 3G was 5.44 times higher for Vodafone and 4.42 times higher for Tim, the two main Italian mobile operators (OpenSignal, 2017). Based on the same report, also latency time was lower under 4G vis a vis 3G in 2017. It was lower by 38.5% for Vodafone and by 43.1% for Tim. Both measures are relevant for assessing the quality of mobile data usage. The fact that the performance of both factors was much higher under 4G should suggest that users were probably more inclined to frequently access social media in the areas with 4G coverage.

$messages_{p,j,t}$ report the total number of messages in a given week for topic j by party p. Outcomes y are poll dummies for Lega, PD, Forza Italia, and Movimento 5 Stelle, and turnout intentions (the share being sure to turn out). Fixed effects are represented by δ_l (location) and δ_t (week). The vector of controls X_i includes age, gender, education, and occupation controls.[22] As in a RDD, we also control for interactions between our communication and events variables and distance from the closest municipality with 4G, as well as for the triple interaction between events or communication, coverage, and distance. Our estimation then identifies the differential effect of events and communication at the border between covered and uncovered areas. For robustness, we also show results without including the distance interactions.

To make sure that the 'treatment' and 'control' locations are comparable, we only use data from areas sufficiently close to the border between 3G and 4G coverage. We choose a benchmark of 20KM, allowing for sufficient variation on both sides of the 4G-coverage border, and experiment with a threshold of 12.5KM for robustness. Since 4G coverage is not measured with precision between 1 and 23 squared kilometers (Manacorda et al., 2022), we also leave out an area closer to the border than 5KM. We cluster errors at the municipality-week level.

In Table 5, we show means and standard deviations, for all of the sociodemographic covariates in the whole sample and in the bandwidth that we have in our analysis. Sample characteristics in the sample used in the RDD analysis are virtually identical to the ones of the overall sample, supporting external validity of our analysis.

As for internal validity, the main identifying assumption for our empirical strategy is that time-varying unobservable characteristics correlated with political outcomes do not change discontinuously at the 4G border. In the last three columns, we use a simple RDD test, with the bandwidth we use in our main analysis, to show that the main observable characteristics do not change at the 4G border. Out of 13 sociodemographic covariates tested, only one occupational category–being in search of the first job–jumps at the threshold. Overall, these results support the external and internal validity of our estimates.

Our main results are reported in Table 6. The first five columns report the estimates of Equation 3 on intentions to vote for Lega, PD, Forza Italia, Movimento 5 Stelle, and for turnout intentions. The estimates on the interaction between Lega's immigration messages and 4G coverage is positive and significant for Lega's polls, and negative and significant for Movimento 5 Stelle's polls. In terms of magnitude, an increase in messages by Lega equal to 0.1-SD in the whole sample of messages by all parties differentially increases voting by Lega by 1.86 percentage points in 4G areas, and reduces votes for Movimento 5 Stelle by 3.06 percentage points.

[22] We include age and age squared, proportion of persons with secondary and tertiary education, and proportion of population in different occupations–self-employed, permanent job, temporary job, apprenticeship, seasonal/occasional job, student, home production, retired, in search of first job, other (all compare to unemployed).

In the last rows of the table we perform two additional tests. In the first, we report the F-stat computed under the null hypothesis that the sum of party's communication coefficients across communication for the column's party is equal to 0. In the second row we report the F-statistic computed under the null hypothesis that no communication category is different from 0 for the column's party; in the turnout column we include all parties' communication variables. The results of the first test suggest that only Lega's communication has a robust effect on own support, while the second test supports the idea that also the communication of Movimento 5 Stelle affects own support in the electorate.

To further explore the connection between exposure to political messages online and intentions to vote, we exploit data from the 2012 wave of the European Social Survey on internet usage to create an indicator for high internet usage predicted by observable characteristics. In particular, we use lasso linear regression to select predictors of daily internet usage among education, occupation, sex, age, age squared, and two-way interactions between them. The resulting index represents a predicted probability of daily internet usage. Using such predictor, we now run our estimation only using data on individuals with high predicted internet usage. Restricting to this sub-sample, the effect of Lega's communication on own polling remains significant and almost doubles while the effect of Lega's communication on Moviemento 5 Stelle's support become much more noisy. The F-stat for the sum of coefficients and for joint significance are similar to the previous version. However, we do not see an effect of Movimento 5 Stelle's communication now, and instead register a less robust effect of Forza Italia's communication, statistically significant at a 10% level.

In Appendix Table D6, we examine the robustness of our estimates. In the first five columns, we drop distance interactions. The effect remains significant for Lega's immigration communication, but much smaller in magnitude. In addition, we now see a negative effect of Lega's communication on the economy on Forza Italia's poll and on turnout. In the last five columns, we experiment with an alternative window, restricted to the right at 12.5KM. The effect of Lega's communication remains significant and is similar to the one computed on the sample of high internet users. Now, F-stat for the sum and the joint significance of coefficients are both statistically significant, together with the F-stat on joint significance of Movimento 5 Stelle's communication.

5 Structural Model

In the previous section, we have shown that populist parties in Italy used immigration events in conjunction with social media communication to increase their support in the electorate. While reduced-form methods are able to establish a causal relationship between events, salience, and

communication, measuring the empirical relevance of such channels in explaining the support for populist parties requires a theoretical framework. In this section, we develop a simple model where a party sets the level of communication on social media over time to influence salience of the preferred topic. We then use the model to quantify the relevance of communication and events in explaining anti-migration parties' support in the electorate.

In the model, voters perceive parties to have a given quality level on each topic $Q_{i,p} \in \{-1/2, 1/2\}$, $i \in \{m, n\}$, modeling either horizontal differentiation–Qs representing the difference in parties' positions and voters' ideology–or vertical differentiation–Qs representing the quality of parties' platform. A voter's expected utility from voting party d is given by the quality of a party's proposed policy on each issue, weighted by the salience of the issue, with γ_t indicating the salience of immigration at time t. This modeling framework captures the idea that voters pay more attention to some topics in choosing their favorite parties, and the aggregate movement of salience is a measure of such level of attention. Finally, voters' utility from choosing a party depends on an additive random component, $\varepsilon_{p,t}$, modeling voters' heterogeneity. We define $\nu_t = \varepsilon_{A,t} - \varepsilon_{B,t}$ and assume that $\mathbb{E}(\nu_t) = 0$, without loss of generality, since the $Q_{p,m}$ values absorb average differences in the relative quality of parties. To be able to write closed-form solutions, we assume more specifically that a uniform distribution in $[-1/2, 1/2]$. Then, a voter will choose to vote party if:

$$U_{p,t} = \underbrace{(\gamma(\mathbf{k}_t^m, \mathbf{k}_t^n, N_t^m, N_t^n) * Q_{p,m} + (1 - \gamma(\mathbf{k}_t^m, \mathbf{k}_t^n, N_t^m, N_t^n)) * Q_{p,n})}_{\text{Party quality weighted by dimension salience}} + \underbrace{\varepsilon_{p,t}}_{\text{Shock}},$$

where the $\Delta Q_j = Q_{j,A} - Q_{j,B}$ represents the difference in the average quality of parties. We also allow for a fraction θ_B of voters that do not respond to platform quality and always votes for the anti-migration party.

To affect the relative salience of each topic, measured by the fraction of searches devoted to migration, parties can set a level of communication $(k_{t,p}^m, k_{t,p}^n) \in \{0, 1\}^2$, modeling a binary choice between high and low communication. The cost of high communication is heterogeneous by party, so that the overall cost of communicating in a given topic is given by $c_p \max\{k_{t,p}^m, k_{t,p}^n\}$. We assume that immigration salience depends on communication and the occurrence of news events based on the following flexible functional form:

$$\begin{aligned}\gamma(\mathbf{k}_t^m, \mathbf{k}_t^n, N_t^m, N_t^n, \nu_t) = (1 + \exp(\alpha_0 + \alpha_1 \max\{k_{t,A}^m, k_{t,B}^m\} + \alpha_2 \max\{k_{t,A}^n, k_{t,B}^n\} + \\ + \sum_j \alpha_j^m N_{m,t}^j + \sum_j \alpha_j^n N_{n,t}^j + \varepsilon_{\gamma,t}))^{-1}\end{aligned}$$

$N_{m,t}$ and $N_{n,t}$ are indicators of the occurrence of news about migration the economy $\varepsilon_{\gamma,t}$ is

governed by an AR process with autocorrelation ρ_γ and standard deviation σ_γ where communication variables in bold represent vectors with dimensions corresponding to parties. The process for the formation of salience allows for a random component as well. The variable $\varepsilon_{\gamma,t}$ represents a shock to the relative attention of immigration, known to the parties, and it is autocorrelated. We define its persistence as ρ_γ, and the standard deviation of the residuals as σ_γ.

Events occurring in our model can affect the relative value of parties' platforms in the eyes of voters, and complement parties' communication's influence on salience. To model this process we assume that the relative advantage of party A in a given topic is given by the following functions of events occurrence dummies,

$$\begin{cases} \Delta Q_{m,t} = \exp\left(\eta_0^m + \sum_j \eta_j^m N_{m,t}^j + \zeta_t^m\right) \\ \Delta Q_{n,t} = \exp\left(\eta_0^n + \sum_j \eta_j^n N_{n,t}^n + \zeta_t^n\right) \end{cases} \tag{4}$$

where $N_{m,t}^j$ is an indicator variable for news about immigration, in subtopic j, occurring in period t, and $N_{n,t}$ is an indicator variable for news about the economy, in subtopic j, occurring in the same period. We allow for two subtopics for immigration–border crossings and first reception versus events involving migrants in Europe–and two subtopics for the economy–corporate and finances events versus state of the economy and redistribution. The variables ζ_t^m and ζ_t^m represent shocks to the advantage of party A on migration and the economy, normally-distributed with mean zero and standard deviations σ_t^m and σ_t^n.

Parties take decisions on how much to communicate and which topics to cover. Their joint decisions, together with the exogenous occurrence of news events, determine the salience of immigration in the economy, which in turn results in the support for the party. We assume that, at every period t, parties play the Nash Equilibrium of this game. An equilibrium is defined as a pair of vectors $(\mathbf{k}_t^{m*}, \mathbf{k}_t^{n*})$ such that, for each party p, the chosen communication levels maximize support in the electorate given the levels chosen by the other party:

$$\begin{aligned} k_t^{m*}, k_t^{n*} = \underset{r_{p,t}^*, k_{p,t}^*}{\operatorname{argmax}} \quad & [\gamma(\mathbf{k}_t^m, \mathbf{k}_t^n, \mathbf{N}_t, \nu_t) * (Q_{p,m} - Q_{-p,m}) + \\ & + (1 - \gamma(\mathbf{k}_t^m, \mathbf{k}_t^n, \mathbf{N}_t, \nu_t)) * (Q_{p,n} - Q_{-p,n})] + \\ & + \varepsilon_{p,t} - \varepsilon_{-p,t} - c_p \max\{k_{t,p}^m, k_{t,p}^n\} \end{aligned} \tag{5}$$

We calibrate the model using a Non-Linear-Least-Squares estimator. Namely, we estimate the model's parameters by choosing the vector of parameters minimizing a distance between the realized communication, polling, and salience and the one simulated by our model. For every parameter draw in the minimization algorithm, we compute all potential Nash Equilibria using

Table 1: Identifying variation

Parameters	Moments
η_0^m, η_0^n, θ_B	support for coalitions, responsiveness of communication to events
ηs	cross-correlation between events, salience and parties' support
c_A, c_B, σ^m, σ^n	mean levels of communication, across topics, variance in parties' support
αs	correlation between events, communication, and searches
σ_γ, ρ_γ	autocorrelation and variance of searches

the Vertex Enumeration algorithm.[23] After computing such equilibria, we simulate the model over the sample period based on a draw of attention shocks, and the realized news events.[24]

We calibrate the model for the years 2015-2018 leading up to the general election of March 2018. Although the political system at the time of the 2018 election was at least partially proportional and clear-cut coalitions were absent, we establish party coalitions that broadly correspond to the right-left political division: we group Lega and Forza Italia, and PD and Movimento 5 Stelle. As for right-wing parties, traditionally, they have had the strongest anti-immigration stances in Italy. The assumption made about Movimento 5 Stelle and PD are somehow less important given the very low level of communication by PD shown in the previous sections.

Before proceeding to the exposition of the results of the model, it is useful to point out which variation in the data intuitively pins down each variable or group of variable, as briefly summarized in Table 1. The mean support for coalitions and their correlation with events intutively pin down η_0^m, η_0^n, and the fraction of voters who are not influenced by salience, θ_B. The cross-correlation between events, salience, and parties support pin down the remaining η parameters affecting the reaction of party advantages to events. Given the effectiveness of communication, mean levels of communication on the issue across topics pin down costs c_A and c_B and the standard deviation of the political preferences shocks σ_t^m and σ_t^n. The mean level of Google searches of migration and their correlation with communication and salience help identify the αs mapping communication and events into the salience of immigration. The correlation between between events and communication and between events of different types and support for parties identify the parameters η_0^k, η_1^k, and η_2^k for topic k. The autocorrelation and variance of searches identify σ_a and ρ_a, determining the autocorrelation of attention. The model and data moments are shown in Table 7. We report the estimated parameters in the Appendix Table D12.

[23] We use the implementation provided by the NashPy module in Python.

[24] To avoid initial conditions in the dataset from polluting our estimates, we drop from the computation of the moments the first five weeks in the sample.

5.1 Counterfactuals

We use the calibrated model to examine two counterfactual scenarios. First, we simulate a counterfactual scenario where we set the probability of immigration events occurring to 0. Under this scenario, we observe a drop in Under this scenario, the simulated vote for the right-wing coalition is about 17% lower than the 24% polling predicted by estimated parameters. Such reduction is caused by a 19% reudction in the salience of migration, as proxied by Google searches, amplified by a 60% decrease in migration communication.

In the second counterfactual scenario, we set the $\eta_1 = 0$, implying that parties are not able to affect the salience of migration through mgiration communication. In this case, the counterfactual support for anti-migration parties would fall by 19%. Such reduction would be caused by a decrease in the salience of migration by 90%. These results are in line with the idea that shifts in exogenous events are not enough to explain the rise of the anti-immigration right, which has also exploited these events using political communication.

6 Conclusions

This paper examines how politicians in Italy strategically use news events and political communication to increase their support in the electorate. We examine the link between news events, online political communication, the salience of different cleavages, and party support. We compare the communication used by populist anti-immigration politicians to the communication of traditional parties, and examine its impact on the support for anti-immigration populism.

We show that parties differ in the response of their daily communication to different types of events, and that immigration events have more pronounced effects on the communication of the populist anti-immigration party. We also show that there is an immediate impact of parties' communication on the salience of different cleavages in the electorate and that this has a causal impact on political support. In particular, we find that communication on immigration increases support for the right-wing populist Lega party.

We then uses a structural model of political competition informed by the relationships uncovered in our reduced-form analysis to evaluate two counterfactual scenarios, one where political parties compete in the absence of communication events, and a second where parties cannot affect migration salience with communication. We find that communication about immigration is more important than immigration events for explaining increased support for the anti-immigration coalition. In the period leading up to the election of 2018, communication explains about 6.9% of the support for the coalition favoring anti-immigration policies, while events only explain 5.6% of the support.

References

Allcott, H. and M. Gentzkow (2017). Social media and fake news in the 2016 election. *Journal of economic perspectives 31*(2), 211–236.

Bianchi, F., D. Nozza, D. Hovy, et al. (2021). Feel-it: Emotion and sentiment classification for the italian language. In *Proceedings of the eleventh workshop on computational approaches to subjectivity, sentiment and social media analysis*. Association for Computational Linguistics.

Bidwell, K., K. Casey, and R. Glennerster (2020). Debates: Voting and expenditure responses to political communication. *Journal of Political Economy 128*(8), 2880–2924.

Bonomi, G., N. Gennaioli, and G. Tabellini (2021). Identity, beliefs, and political conflict. *The Quarterly Journal of Economics 136*(4), 2371–2411.

Bordalo, P., N. Gennaioli, and A. Shleifer (2012). Salience theory of choice under risk. *The Quarterly journal of economics 127*(3), 1243–1285.

Campante, F., R. Durante, and F. Sobbrio (2018). Politics 2.0: The multifaceted effect of broadband internet on political participation. *Journal of the European Economic Association 16*(4), 1094–1136.

Dustmann, C., K. Vasiljeva, and A. Piil Damm (2019). Refugee migration and electoral outcomes. *The Review of Economic Studies 86*(5), 2035–2091.

Enikolopov, R., A. Makarin, and M. Petrova (2020). Social media and protest participation: Evidence from russia. *Econometrica 88*(4), 1479–1514.

Fujiwara, T., K. Müller, and C. Schwarz (2021). The effect of social media on elections: Evidence from the united states. Technical report, National Bureau of Economic Research.

Galasso, V., T. Nannicini, and S. Nunnari (2023). Positive spillovers from negative campaigning. *American Journal of Political Science 67*(1), 5–21.

Giannetti, D., A. Pedrazzani, and L. Pinto (2017). Party system change in italy: politicising the eu and the rise of eccentric parties. *South European Society and Politics 22*(1), 21–42.

Hangartner, D., E. Dinas, M. Marbach, K. Matakos, and D. Xefteris (2019). Does exposure to the refugee crisis make natives more hostile? *American political science review 113*(2), 442–455.

Ignazi, P., E. Risso, and S. Wellhofer (2022). *Elezioni e partiti nell'Italia repubblicana*. Società editrice il Mulino, Spa.

Informa UK (2013). Understanding today's smartphone user: an updated and expanded analysis of data-usage patterns in six of the world's most advanced 4G markets. https://www.derekspratt.com/PDFs/Business/Mobidia/Informa%20Report%20-%20An%20Updated%20and%20Expanded%20View%20of%20LTE%20Usage%20-%20June%202013.pdf.

Informa UK (2014). Smartphone use transforming with the rise of 4G and Wi-Fi. https://www.derekspratt.com/PDFs/Business/Mobidia/Informa%20Report%20-%20Smartphone%20Use%20Transforming%20with%20the%20Rise%20of%204G%20and%20Wi-Fi%20-%20Feb%202014.pdf.

Kendall, C., T. Nannicini, and F. Trebbi (2015). How do voters respond to information? evidence from a randomized campaign. *American Economic Review 105*(1), 322–353.

Le Pennec, C. and V. Pons (2023). How do campaigns shape vote choice? multicountry evidence from 62 elections and 56 tv debates. *The Quarterly Journal of Economics 138*(2), 703–767.

Levi, E., R. D. Mariani, and M. Mongiardo (2019). *L'ostilità verso l'immigrazione: la percezione del fenomeno migratorio: i fallimenti della politica, il framing mediatico e la socialdemocrazia europea*. Rubbettino Editore.

Levy, R. (2021). Social media, news consumption, and polarization: Evidence from a field experiment. *American economic review 111*(3), 831–870.

Manacorda, M., G. Tabellini, and A. Tesei (2022). Mobile internet and the rise of political tribalism in europe.

Müller, K. and C. Schwarz (2023). From hashtag to hate crime: Twitter and antiminority sentiment. *American Economic Journal: Applied Economics 15*(3), 270–312.

OpenSignal (2017). State of mobile networks: Italy. https://www.opensignal.com/reports/2017/05/italy/state-of-the-mobile-network.

Pasquino, G. and M. Valbruzzi (2017). The italian democratic party, its nature and its secretary. *Revista Española de Ciencia Política* (44), 275–299.

Piskorski, J., H. Tanev, M. Atkinson, and E. Van Der Goot (2008). Cluster-centric approach to news event extraction. *New Trends in Multimedia and Network Information Systems 181*, 276.

Spenkuch, J. L. and D. Toniatti (2018). Political advertising and election results. *The Quarterly Journal of Economics 133*(4), 1981–2036.

Steinmayr, A. (2021). Contact versus exposure: Refugee presence and voting for the far right. *Review of Economics and Statistics 103*(2), 310–327.

Tronconi, F. (2020). The italian five star movement during the crisis: towards normalisation? *Italy Transformed*, 162–179.

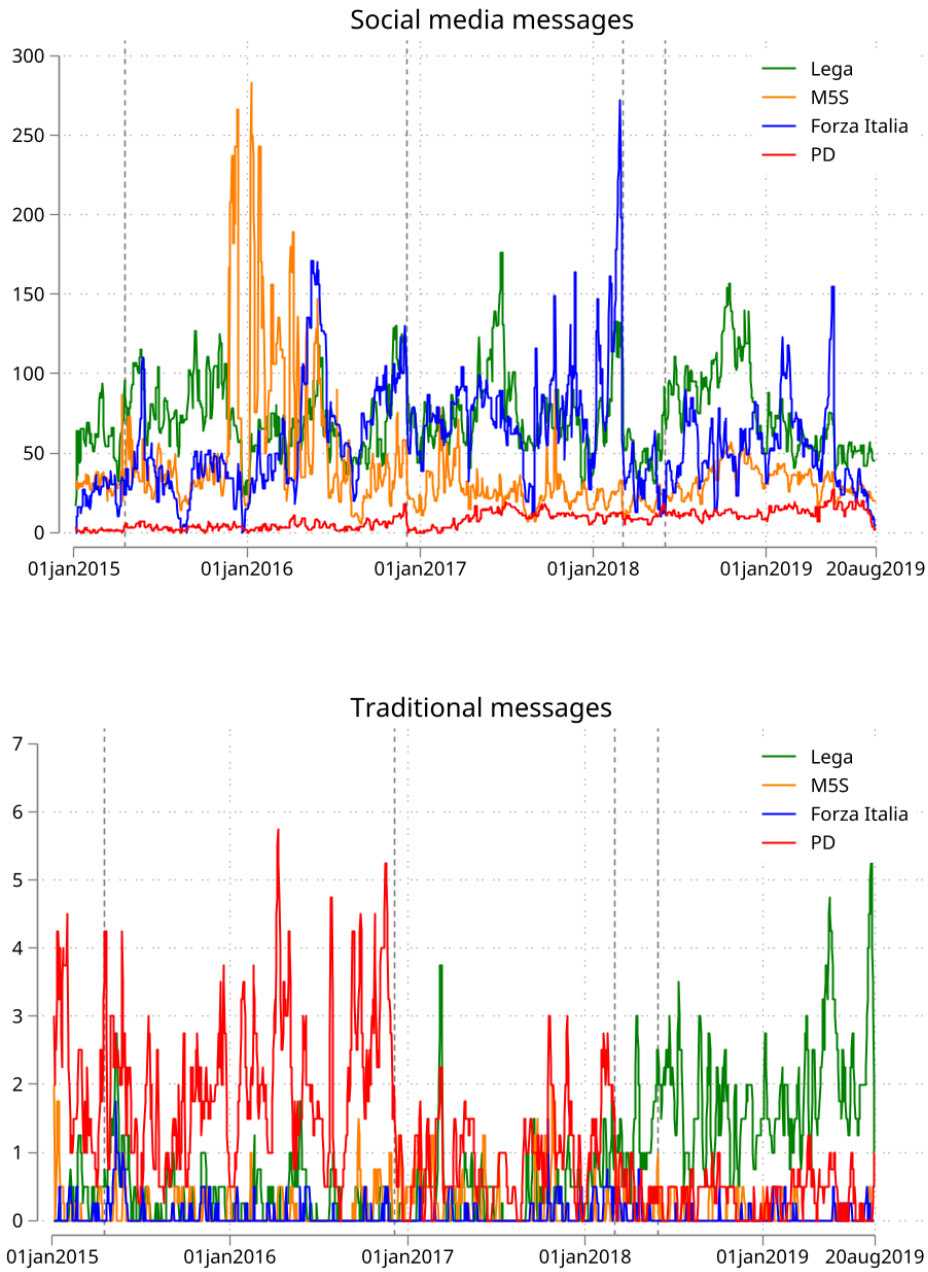
Vassallo, S. and M. Valbruzzi (2018). I partiti della grande recessione contro i partiti della grande depressione. un nuovo cleavage o un'altra bolla? *Stato e mercato 38*(1), 87–116.

Table 2: Impact of News on Social Media Usage by Political Parties

	Leftwing				*Rightwing*			
	Social media		*Traditional media*		*Social media*		*Traditional media*	
Populist parties	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	M5S on Imm	M5S on Eco	M5S on Imm	M5S on Eco	Lega on Imm	Lega on Eco	Lega on Imm	Lega on Eco
Economic events	0.000	0.007	-0.000	-0.002	-0.002	0.012***	0.004	0.009
	(0.003)	(0.005)	(0.000)	(0.004)	(0.004)	(0.003)	(0.007)	(0.007)
Immigration events	0.001	0.008***	0.000	0.002	0.027***	-0.000	0.016***	0.001
	(0.002)	(0.003)	(0.000)	(0.002)	(0.003)	(0.002)	(0.005)	(0.005)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.095	0.230	0.041	0.058	0.341	0.267	0.132	0.132
Traditional parties	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	PD on Imm	PD on Eco	PD on Imm	PD on Eco	FI on Imm	FI on Eco	FI on Imm	FI on Eco
Economic events	-0.001**	0.007***	0.008**	0.034***	0.002	0.030***	0.000	-0.000
	(0.000)	(0.001)	(0.004)	(0.013)	(0.002)	(0.007)	(0.000)	(0.002)
Immigration events	0.002***	-0.001	0.009***	0.004	0.007***	-0.005*	0.000	-0.001
	(0.000)	(0.001)	(0.003)	(0.005)	(0.001)	(0.003)	(0.000)	(0.001)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.159	0.475	0.080	0.143	0.363	0.437	-6.643	0.049

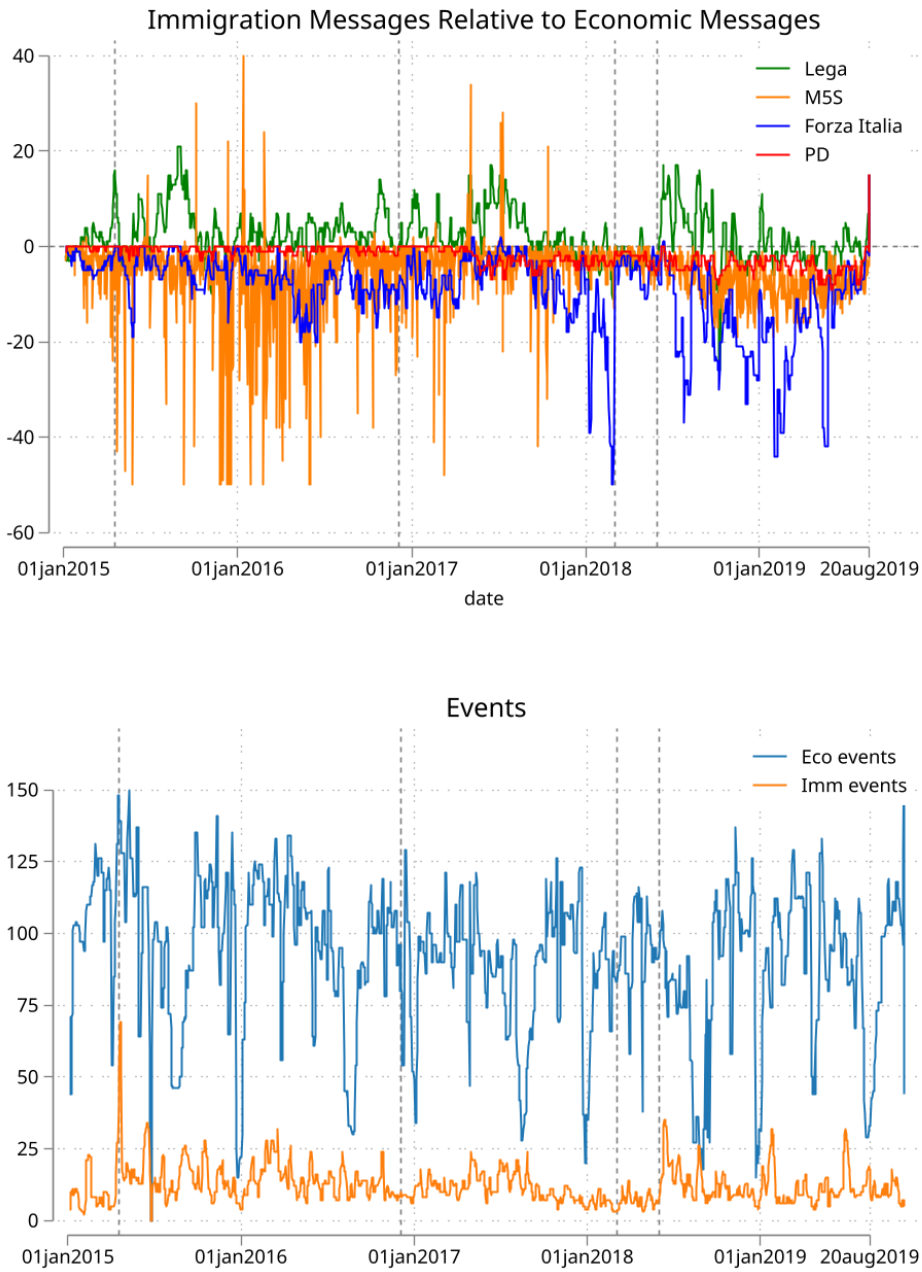
Notes: Each column corresponds to an OLS regression. All variables on events are standardized to be mean zero and standard deviation one across the sample. All variables on messages are standardized with respect to the total number of social media or traditional messages. All regressions include two lags of the total number of messages, day of week, week of year, and year fixed effects as well as daily time trends that vary by year. SEs robust to arbitrary heteroskedasticity and within-panel autocorrelation and cross-panel autocorrelated disturbances that disappear after 6 lags. *, **, ***, indicate statistical significance at the 10%, 5%, and 1% level.

Figure 1: Social media and traditional messages over time



Social media and traditional messages over time. Dashed lines represent events of major political relevance in Italy: in this order, the start of the 2015 refugee crisis, the 2016 referendum on constitutional reforms, the 2018 general elections, and the formation of the Conte I government. The time series on social media messages is smoothed using a non linear filter with a median smoother of span 7, the one on traditional messages with a median smoother of span 28.

Figure 2: Messages and events over time



On the top panel, the difference between immigration and economic messages over time. On the bottom panel, events over time. Dashed lines represent events of major political relevance in Italy: in this order, the start of the 2015 refugee crisis, the 2016 referendum on constitutional reforms, the 2018 general elections, and the formation of the Conte I government. The time series is smoothed using a non linear filter with a median smoother of span 7.

Table 3: Impacts of social media messages by Political Parties on the difference in Hourly Google Searches between immigration and economic topics

	Imm-Eco Google searches			
	(1)	(2)	(3)	(4) - LDA
On imm	0.1680***	0.1383**		
	(0.0572)	(0.0578)		
On eco	-0.0537	-0.0494		
	(0.0353)	(0.0361)		
M5S on imm			-0.0515	-0.0927
			(0.1190)	(0.1194)
M5S on eco			-0.0272	-0.0244
			(0.0584)	(0.0587)
Lega on imm			0.2389***	0.2563***
			(0.0787)	(0.0788)
Lega on eco			-0.1719*	-0.2024**
			(0.0931)	(0.0936)
PD on imm			1.4340*	1.8610**
			(0.8268)	(0.8322)
PD on eco			-0.1443	-0.1005
			(0.2047)	(0.2068)
FI on imm			-0.0314	0.0128
			(0.1625)	(0.1629)
FI on eco			0.0028	-0.0016
			(0.0631)	(0.0630)
Immigration events		0.0446***	0.0438***	
		(0.0078)	(0.0079)	
Economic events		-0.0192**	-0.0182**	
		(0.0078)	(0.0080)	
F-Stat M5S across types			0.66	1.07
F-Stat Lega across types			4.76***	5.47***
F-Stat PD across types			1.46	2.17*
F-Stat FI across types			0.14	0.44
N	40316	40316	40316	40316
r2	0.1118	0.1125	0.1129	0.1137

Notes: Searches for the Immigration and Economic topics are hourly searches from Google Trends for the 10 most prominent words based on importance in the machine learning classification task. The dependent variable measures standardized differences between searches on Immigration and Economic topics. RHS variables on events and messages include all total number of traditional or social media messages in the previous 24 hours. All RHS variables on messages are standardized with respect to the total number of social (traditional) media messages. RHS variables on events are standardized with respect to their mean and standard deviation. We control in all regressions for social media messages on other topics and for traditional media messages for each party and in each category. All regressions include 196 hour lags (1 week) of Google Trends searches, date and hour of the day fixed effects. Regressions in last column includes controls for events on immigration and the economy by the type of event such as classified by a Latent Dirichlet Allocation over 14 categories (more details in the text). SEs robust to arbitrary heteroskedasticity and within-panel autocorrelation and cross-panel autocorrelated disturbances that disappear after 14 lags. *. **, ***, indicate statistical significance at the 10%, 5%, and 1% level.

Table 4: Impacts of social media messages by Political Parties on the difference in Hourly Google Searches between immigration and economic topics - Robustness

	12 hours	20 keywords	No events
M5S on imm	-0.0903	-0.0002	-4.8706*
	(0.1116)	(0.1271)	(2.8627)
M5S on eco	-0.0660	-0.0556	1.2854
	(0.0567)	(0.0624)	(3.6566)
Lega on imm	0.3066***	0.2433***	7.4145**
	(0.0693)	(0.0837)	(3.3296)
Lega on eco	-0.1443*	-0.2070**	-8.5084**
	(0.0817)	(0.0996)	(3.3840)
PD on imm	1.2978*	2.3156***	4.0234
	(0.7823)	(0.8851)	(15.0059)
PD on eco	0.0819	-0.2737	-2.8448
	(0.1912)	(0.2201)	(12.2723)
FI on imm	-0.0338	-0.0608	24.8478
	(0.1430)	(0.1734)	(15.4119)
FI on eco	-0.0262	-0.0230	-1.5955
	(0.0542)	(0.0670)	(1.8426)
N	40316	40316	91
r2	0.1142	0.1010	0.5261

Notes: The main specification is as follows, then each column deviates from this specification in one respect, listed in the header and at the end of the notes. Searches for the Immigration and Economic topics are hourly searches from Google Trends for the 10 most prominent words based on importance in the machine learning classification task. The dependent variable measures standardized differences between searches on Immigration and Economic topics. RHS variables on events and messages include all total number of traditional or social media messages in the previous 24 hours. All RHS variables on messages are standardized with respect to the total number of social (traditional) media messages. RHS variables on events are standardized with respect to their mean and standard deviation. We control in all regressions for social media messages on other topics and for traditional media messages for each party and in each category. All regressions include 196 hour lags (1 week) of Google Trends searches, date and hour of the day fixed effects. Regressions includes controls for events on immigration and the economy by the type of event such as classified by a Latent Dirichlet Allocation over 14 categories (more details in the text). SEs robust to arbitrary heteroskedasticity and within-panel autocorrelation and cross-panel autocorrelated disturbances that disappear after 14 lags. In column (1) the RHS variables encompass lags up to 12 hours rather than 24. In column (2) searches are for the 20 most prominent words based on importance in the machine learning classification task (see appendix). In column (3) we only consider hours without any event, with 12 hours lags of the dependent variable and no time fixed effects. *, **, ***, indicate statistical significance at the 10%, 5%, and 1% level.

Table 5: Differences in covariates around the 4G border

	Mean in all sample	Mean for observations w/in bandwidth	4G Coverage	Distance from 4G Border	4G Coverage X Distance from 4G Border
Age	50.018	50.029	-1.387	-0.037	-0.079
	(16.148)	(16.257)	(1.224)	(0.045)	(0.115)
Female	0.524	0.521	0.015	0.001	0.000
	(0.499)	(0.500)	(0.023)	(0.001)	(0.002)
Second. education	0.735	0.730	-0.018	-0.003**	0.008***
	(0.441)	(0.444)	(0.030)	(0.001)	(0.003)
Tert. education	0.205	0.212	0.026	0.003**	-0.009***
	(0.404)	(0.409)	(0.030)	(0.001)	(0.003)
Self-employed	0.101	0.100	-0.013	-0.000	-0.001
	(0.301)	(0.300)	(0.014)	(0.001)	(0.001)
Permanent	0.277	0.279	0.033	0.001	0.002
	(0.447)	(0.449)	(0.024)	(0.001)	(0.002)
Temporary	0.048	0.049	-0.002	0.000	-0.001
	(0.213)	(0.215)	(0.008)	(0.000)	(0.001)
Apprenticeship	0.006	0.006	-0.001	-0.000	-0.000
	(0.078)	(0.079)	(0.002)	(0.000)	(0.000)
Seasonal/occasional	0.009	0.010	-0.002	0.000	-0.000
	(0.097)	(0.097)	(0.004)	(0.000)	(0.000)
Student	0.052	0.051	0.008	0.001**	-0.002**
	(0.222)	(0.220)	(0.009)	(0.000)	(0.001)
Home production	0.152	0.151	0.017	-0.001	0.005
	(0.359)	(0.358)	(0.030)	(0.001)	(0.003)
Retired	0.242	0.245	-0.061	-0.002	-0.003
	(0.428)	(0.430)	(0.039)	(0.001)	(0.004)
In search of first job	0.016	0.017	0.014**	0.001**	-0.000
	(0.127)	(0.130)	(0.006)	(0.000)	(0.001)
Other occupation	0.007	0.006	0.002	0.000	0.000
	(0.082)	(0.078)	(0.003)	(0.000)	(0.000)

Notes: This table shows the coefficients of the RDD regressions using distance from the 4G border as the running variable, employing the window used in the main analysis, with a minimum distance threshold of 5KM, and a maximum of 20KM, on all socio-demographic observables included in the analysis. Each line reports coefficient and standard errors for a regression on one observable. Standard errors are clustered at the Comune level. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table 6: Impacts of Social media messages Political Parties on Voting and Turnout

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
						High internet users				
	5KM < distance < 20KM					5KM < distance < 20KM				
	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout
4G Coverage X Past week's econ events	-0.00826	0.00300	0.00684	0.0119	0.0146	-0.0104	0.0286	0.00470	0.0282	0.0379
	(0.00955)	(0.0126)	(0.00923)	(0.0125)	(0.0182)	(0.0166)	(0.0202)	(0.0104)	(0.0174)	(0.0256)
4G Coverage X Past week's imm events	0.0101	0.00349	0.00170	-0.0125	0.0206	0.00732	0.00122	0.00618	-0.0156	-0.0112
	(0.00920)	(0.0117)	(0.00811)	(0.0109)	(0.0212)	(0.0127)	(0.0164)	(0.0110)	(0.0173)	(0.0290)
4G Coverage X Past week's imm messages by Lega	0.186**	0.112	-0.0359	-0.306**	0.105	0.371***	0.0517	-0.00691	-0.159	0.282
	(0.0896)	(0.132)	(0.0743)	(0.139)	(0.198)	(0.129)	(0.195)	(0.101)	(0.178)	(0.302)
4G Coverage X Past week's econ messages by Lega	0.124	-0.250	-0.113	0.126	-0.0131	0.152	0.0105	-0.0260	-0.344	0.323
	(0.151)	(0.207)	(0.118)	(0.220)	(0.343)	(0.243)	(0.289)	(0.164)	(0.278)	(0.490)
4G Coverage X Past week's other messages by Lega	0.00916	0.0492	0.0211	0.0359	0.0427	0.0319	0.0331	0.0415	0.0549	0.0357
	(0.0297)	(0.0462)	(0.0279)	(0.0441)	(0.0605)	(0.0518)	(0.0553)	(0.0299)	(0.0560)	(0.0857)
F-Stat of sum of same-party communication across types	4.207**	2.227	0.112	0.050		5.658**	0.729	2.838*	1.439	
F-Stat of all same-party communication across types	2.830**	1.290	0.051	2.443*	0.830	4.093***	0.320	2.285*	1.768	1.632*
N	72601	72601	72601	72601	56986	36519	36519	36519	36519	28968

Notes: This table shows the coefficients of the regression estimating Equation 3 in columns (1-4), with a minimum distance threshold of 5KM, and a maximum of 20KM in columns (1), (2), and of 10KM in columns (3) and (4). Columns (5) and (8) replicate the analysis on voters predicted to be high internet users, based on a model trained on data of the European Social Survey of 2012. All of the estimations include municipality FEs, week FEs, political communication interactions for other parties (Partito Democratico, Forza Italia, Movimento 5 Stelle), two-way and three-way interactions between communication and events variables with 4G coverage and distance from the 4G border. Finally, we control for respondent's socio-demographic characteristics: age, age squared, dummies for education level and occupation. The last rows in the table report (i) the F-stat relative to the null hypothesis that the sum of communication coefficients for the columns' party is equal to 0 and (ii) the F-stat relative to the null that all same-party's communication coefficients are jointly 0. In the latter test, we compute the joint F-stat on all parties when the dependent variable is turnout. Standard errors are robust to heteroskedasticity. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table 7: Moments: data and model simulations

Moment description	Data	Model
Mean of rightwing support	0.204	0.247
Corr. b/w imm. messages and log searches	0.317	0.2
Corr. b/w econ. messages and log searches	0.337	-0.857
Mean imm. communication by rightwing parties	0.511	0.605
Mean econ. communication by rightwing parties	0.856	0.27
Mean imm. communication by leftwing parties	0.007	0.262
Mean econ. communication by leftwing parties	0.144	0.559
Mean log searches	-3.735	-3.914
Corr. between rightwing support	0.061	0.197
Autocorr. of log searcher	0.605	0.566
Standard deviation of log searches	0.605	3.427
Standard deviation of rightwing support	0.002	0.037

A Machine learning classifications

In this appendix we will explain in more detail how we classified social media messages and how we extracted events from news agencies.

Social media messages: the Facebook posts classified with a supervised machine algorithm. So, the first step was having a sample classified by a research assistent. After having stripped the dataset from all information related to the post apart from the actual content of the FB post, we randomly restricted the sample to 10000 messages (2500 by Facebook public page).

The following instructions were given to the research assistant:

Instructions for the human rater of political messages

Attached, you will find an excel file called *sample_facebook_RA*. There are 4 columns in this file: the first ('id') contains an identification code that you can ignore, the second ('message') contains a few lines of text, more precisely a Facebook post from an Italian politician or party, while the others ('category1', 'category2', 'category3') are empty.

What we would ask you is to carefully read the following list, which represents a set of labels, and then label the texts in the excel file using this list. The labels that will be applied will be those that best describe the topics addressed in the text.

If you think the text has only one of the topics presented in the tags in the list, please enter the number corresponding to the tag in the 'category1' column. If you think the text has more than one topic, you can enter up to 2 additional ones in 'category2' and 'category3'. The 'category1' column must be filled in for each row; In this regard, we invite you to use the residual label (19) only if it is not possible to apply any other. For this purpose, the classification index also contains examples, which however should not be considered exhaustive of the cases in question.

It is important that this operation is done with the utmost precision and scrupulousness. In particular, we would ask for particular attention on labels 5 (economy) and 7 (immigration), in the sense of not missing texts that may belong to these two categories.

Politics

1. party appointments (rallies, leafleting, party parties, demonstrations, training schools, online meetings, etc.)

2. media appearances of party members (TV programs, radio, newspapers, interviews)

3. appeals to citizens/voters or declarations on internal political alliances

4. political attacks on opponents

National issues

5. economy (banks, pensions, welfare, GDP, work, investments, unemployment, trade unions, taxation, business incentives, third sector, gender gap, companies, etc.)
6. environment (climate change, hydrogeological instability, green incentives)
7. immigration (refugees, immigrants, arrivals-by-sea, etc.)
8. security (crimes)
9. corruption and legality (law sweepers, mafias, scandals, etc.)
10. justice (wiretapping, justice reforms, political fugitives, etc.)
11. education (school, university)
12. health and disability (doctors, hospitals, reforms of the national health system)
13. institutional issues (electoral law, constitutional reforms, quality of democracy, constitutional referendum, federalism, RAI)
14. civil rights and ethically sensitive issues (LGBT, bioethics, prostitution)
15. purely local issues (Regions, Municipalities, mayors, earthquakes)

International issues

16. Europe and European policies
17. foreign policy (conflicts, international cooperation, international meetings, diplomatic meetings)
18. Islamic terrorism/radical Islamism

Other

19. other
20. anomalies (just an unclassifiable link or topic)

NOTE: 1. Some posts contain links. Please do not open them and rate posts only based on the text contained in the file. 2. There are texts with no words (or only a link): label 20 (anomalies) is the one that will be applied in these cases; 3. label 4 (political attacks on opponents) will be applied to all those texts which explicitly refer in a polemical manner to the actions or statements of political opponents. It should be noted that this label often coexists with thematic labels, because the political opponent is often referred to in the text in relation to a specific topic. For example, a Lega post criticizing the PD in relation to migration policies would fall into categories 4 (attacks on opponents) and 7 (migration). 4. We invite an extensive application of the labels if the possibility of associating the text to more than one category is configured. However, we would ask not to exceed the number of 3 labels for each text, making in the (rare) cases in which it is believed that more than 3 labels, an assessment of the relevance of each label in better describing the topic of the text should be applied.

After receiving the classified sample from the research assistant, we proceeded with the usual text pre-processing. We lowered the case of each character, tokenized the text, removed stopwords, links, special characters and words with 1 character only. We then aggregated all categories except economy and immigration into a single one. The pre-processing was done with the nltk package in python. For the classification, we used the standard sklearn package. After extracting a 35% training sample, we put the data into a machine learning pipeline which first vectorized the text either as unigram or bigram, with a cut-off for the frequency either at 0.10, 0.35, 0.5, or 0.75, and a number of features either of 5000, 10000, or 15000, then normalized it with a tf-idf representation. Then, we considered several possible classifiers in their multi-output version: logistic regression, Multinomial Bayes, Support Vector Machine (SVM), Random Forest. Using multi-output classifiers is relevant, for example, for messages on the economic effect of migration, which belong to both our categories. For each one of the classifiers, we pipelined different combinations of parameters. For the winning classifier, the SVM, we considered several possibilities for the regularization parameter C (namely, 0.9, 0.75, 0.5, 0.35), for the type of kernel (linear, polynomial, RBF) and the decision function (one-vs-rest vs one-vs-one). The weight of each class was always balanced so to account for the highest number of data points in the residual category. The best combination was unigrams, a cut-off of 0.35, a number of features equal to 10000, and a SVM classifier with a C of 0.75, a linear kernel with a one-vs-rest decision function. Based on out-of-the-box prediction on the testing sample, this classifier was able to correctly predict 90.8% of the categories. Additional statistics by category are provided in Table A1. Even with this fine-grained analysis, still the number of false positives and false negatives is under 50% for each category.

	Precision	Recall	F1-score	Support
Economy	0.72	0.72	0.72	218
Immigration	0.53	0.85	0.65	65
Residual	0.93	0.93	0.93	993

Table A1: Performance of the classification on the Facebook testing sample

	Precision	Recall	F1-score	Support
Economy	0.68	0.61	0.64	311
Immigration	0.60	0.80	0.69	139
Residual	0.93	0.94	0.93	1535

Table A2: Performance of the classification on the Twitter testing sample

A similar approach was used for the Twitter tweets. 5000 tweets were randomly extracted from the universe of PD and Lega tweets, 2500 for each party (be them of the leader or of the official party account). Then, a research assistant coded these tweets, stripped of everything except the content of the tweet, into 19 categories. The instructions were the same as for Facebook with minor differences.

After receiving the classified sample from the research assistant, we proceeded with the usual text pre-processing and aggregated all categories except economy and immigration into a single one. After extracting a 35% training sample, we put the data into the same machine learning pipeline as the Facebook data. The same classifiers were also considered with the same combination of parameters and the weight of each class was always balanced. The best combination was bigrams, a cut-off of 0.1, a number of features equal to 15000, and a logistic regression classifier with a C of 0.9 and 3000 maximum iterations. Based on out-of-the-box prediction on the testing sample, this classifier was able to correctly predict 90.2% of the categories. Additional statistics by category are provided in Table A2. Even with this fine-grained analysis, still the number of false positives and false negatives is under 40% for each category.

Events: procedure here is very similar to that of social media messages. The main difference, as already described in the main text, resides in excluding certain types of agencies which would confound our results and/or add up to the noise. Also instructions and starting categories are different (e.g., we asked to focus on the title rather than on the text):

Instructions for the human rater of newswires

Attached, you will find an excel file called 'sample_ANSA_RA'. There are 5 columns in this file: the first ('id') contains an identification code that you can ignore, the second ('text') is hidden

and contains the text of an ANSA press release, the third ('title') contains the title, while the others ('category1', 'category2', 'category3') are empty.

What we ask you is to carefully read the following list, which represents a set of labels, and then to label the texts present in the excel file with the number of categories relating to the news, using this list. The labels to be applied will be those that best describe the events described in the title ("title" column). In the rare cases where the title is not sufficient to identify the topic of the news, we ask you to open the "text" column and use its content for classification. N.B. If the title and/or text of the news does not describe or present events but reports statements by politicians or civil society actors, please use the residual labels (12, 26, 39).

If you think the news relates only to one of the topics presented in the tags in the list, please enter the number corresponding to the tag in the "category1" column. If you think the text has more than one topic, you can enter up to 2 additional ones in "category2" and "category3". The "category1" column must be filled in for each row; In this regard, we invite you to use the residual label only if it is not possible to apply any other. For this purpose, the classification index also contains examples, which however should not be considered exhaustive of the cases in question.

For the classification, we advise you to first determine if the news concerns the macro-categories "immigration", "economy" or "other". After identifying the macro-category, you can scroll through the relative list to find the most appropriate category.

Macrocategory	Category	Code
Immigration	Landings, rescues at sea or shipwrecks of boats with immigrants both in Italy and in the rest of the world	1
	Entry of immigrants from other countries with alternative means to boats (truck, train, etc.)	2
	Movements of immigrants between Italian municipalities and outflows from Italy to other countries by any means and gatherings of immigrants traveling towards the border (e.g. Ventimiglia, Brennero, Milano Centrale railway station, etc.)	3
	Crimes against immigrants, when the perpetrator is Italian, with a racial or political background	4
	Phenomena of illegal hiring or in general of exploitation of immigrant labor for work purposes	5
	Trafficking in human beings, therefore everything concerning immigration movements managed by the underworld or the smuggling phenomenon	6

	Scams or scandals related to reception policies (including some episodes of Mafia Capitale)	7
	All crimes in which at least one of the perpetrators is an immigrant that does not fall under the previous two points	8
	Islamic terrorism	9
	Immigration-themed cultural events, pro-immigration political rallies and local integration events/policies	10
	Political demonstrations and riots in which immigrants are the protagonists	11
	Other immigration news (including statements, NGO reports, international summits, statements by church leaders)	12
Economy	Banks and the financial market, understood as news on bank performance, on scandals involving banks, on public interventions concerning the financial markets and the banking sector	13
	Returns and performance of individual companies (except company crisis)	14
	Pensions and welfare, including the Fornero reform, subsequent corrections, and data on the sustainability of the pension system	15
	Employment, unemployment, including deaths at work and renewals of national labor contracts	16
	Economic growth, investment, business incentives, both official statistics and local news on specific business investment, or incentives to sectors of the economy	17
	Infrastructures, including, for example, school buildings, motorways, the development of high-speed rail, the Tav, etc.	18
	Unions and strikes	19
	Fisco, that is, everything related to changes in the tax system and taxes	20
	Energy, i.e. all information concerning the price of energy, fluctuations in the price of fossil fuels (oil, gas, coal, etc.), geopolitical turmoil related to energy	21
	Corporate crises, including ministerial negotiation tables between businesses and companies	22
	Corporate transactions, i.e. mergers between companies, restructuring of the corporate structure, acquisitions, dividends, etc.	23

	Gender gap, wage inequality, discrimination on the labor market, generational issues, maternity and paternity leave	24
	International trade, including trade fairs and exhibitions (Expo), disputes between USA/Europe and China (Road Belt Initiative), issues concerning international free trade treaties	25
	Other economic news (including statements)	26
Other	Environment (climate change, hydrogeological instability, green incentives, fires, extreme bad weather)	27
	Security (crimes of Italians against Italians or it is not clear whether the perpetrator is an immigrant or not)	28
	Corruption and legality (law sweepers, mafias, scandals, etc.)	29
	Corporate crimes, investigations into companies	30
	Justice (wiretapping, justice reforms, political fugitives, etc.)	31
	Education (school, university)	32
	Health and disability (doctors, hospitals, reforms of the national health system)	33
	Institutional issues (electoral law, constitutional reforms, quality of democracy, constitutional referendum, federalism, RAI)	34
	Civil rights and ethically sensitive issues (LGBT, bioethics, prostitution)	35
	Purely local issues (Regions, Municipalities, mayors, earthquakes)	36
	Europe and European politics	37
	Foreign policy (conflicts, international cooperation, international meetings, diplomatic meetings)	38
	Other (including statements by politicians)	39

Note: We invite an extensive application of the labels if the possibility of associating the text to more than one category is configured. However, we would ask not to exceed the number of 3 labels for each text, making in the (rare) cases in which it is believed that more than 3 labels, an assessment of the relevance of each label in better describing the topic of the text should be applied.

We remain of course available in case of doubts or questions. A thousand thanks!

After receiving the classified sample from the research assistant, we proceeded with the usual text pre-processing and aggregated categories into three ones (economy, immigration, and other).

After extracting a 35% training sample, we put the data into the same machine learning pipeline as the Facebook and Twitter data. We ran the classification both over the text and the title of the agencies while ultimately going for the text because of a slight increase in the performance of the algorithm. The usual classifiers were considered with the usual combination of parameters and the weight of each class was always balanced. The best combination was bigrams, a cut-off of 0.1, a number of features equal to 15000, and a SVM classifier with a C of 0.5 and a linear kernel. Based on out-of-the-box prediction on the testing sample, this classifier was able to correctly predict 85.7% of the categories. Additional statistics by category are provided in Table A4. Even with this fine-grained analysis, still the number of false positives and false negatives is under 30% for each category.

	Precision	Recall	F1-score	Support
Economy	0.71	0.83	0.77	195
Immigration	0.85	0.77	0.81	1348
Residual	0.87	0.90	0.89	2477

Table A4: Performance of the classification on the ANSA testing sample

B Feature importance

Choosing the right keywords for our Google Trends queries is of paramount importance. We need keywords that are representative of the topic and avoid choosing arbitrary ones based on our impressions. This is why we decided to go for an unsupervised method. More specifically, we exploited our machine learning classification of ANSA articles. By unboxing the SVM results, we extracted the feature importance of each word. Feature importance indicates how much each feature contributes to the model prediction. To say it in other words, it is the (positive) coefficient attached to each feature, i.e. to the name of each variable. Figure B1 has the names of the 30 top features together with their coefficients. Most economic names relate to economic institutions (e.g. banks, workers, firms, markets) or activities (expo, manovra (policy), growth) with coefficients that range between 2.5 and 1. The migration topic is well predicted by the name "migrants" alone, as the coefficient is above 7, however other words are also relevant (terrorism, immigration, refugees, Syria, asylum, Lybia, hospitality, etc.). Overall, results are reassuring, both for the quality of the classification and for the possibility of using these names for Google Trends queries. In the main results, we consider Google Trends series built from the first 10 of these words to avoid considering names of brands for the Google searches on the economy, but we also tried with all 20 keywords in robustness tests in Table 4.

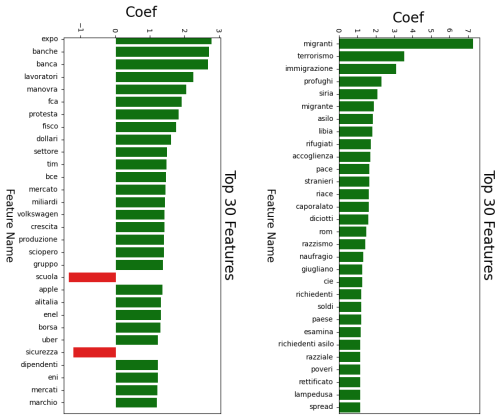


Figure B1: Feature importance for the Economic and Immigration topics

C Type of events

To look at the impact of specific events, we perfomed a Latent Dirichlet allocation (LDA) over the ANSA agencies within the immigration and the economic category separately. The LDA is an unsupervised model for topic classification, which is very helpful in discovering the "hidden" topics in a collection of documents. Basically, the only inputs are the collection of documents, i.e. the ANSA agencies, and a pre-defined number of topics. Then, the model assigns a probability to each word that it belongs to a certain topic based on frequency of co-occurring with other words. After several tries, we opted for 14 categories, as with this option the keywords associated with the higher probability to the categories created consistent categories that we then labeled. Choosing a different number of categories delivered spurious categories that were hard to identify. We represent the topics and their associated most frequent words in Figure C1 and C2. The LDA, by computing a weighted sum of probabilities over each word in each agency, left us with a distribution of probability over topics associated to each agency. We used these probabilities to inform a regression model in which each topic is associated to the agencies with a certain value, i.e. the probability that the agency pertains to that topic. In that way, we can understand if political online communication on the economy or on immigration is driven by specific immigration or economy events. Results in Figure C3 indicate that this is indeed the case. Lega's penchant for immigration over the economy is driven by hospitality, arrivals-by-sea, policy-related issues, rescues and international trade. This is consistent with Lega's focus on illegal migrants. The lack of focus on the economy is well depicted by the absence of a strategic use of specific events on the economy; we do find weak effects on international trade consistent with a mild protectionist stance, but nothing more. For the other parties, we find evidence that

they care more about the economy, even though each party with its own focus: M5S on migrants' exploitation, EU rules, media and new technologies, FI on firms, energy and communication, and labour and pensions, PD on firms and labour and pensions. Then, when Renzi was Prime Minister, a diplomatic case with Austria exploded over migrants' illegally passing the border: it is not surprising that PD tried to deflect the issue by speaking more on the economy. These differences well reflect the different focus of these parties when it comes to economic matters: M5S used to be an anti-austerity party promoting a reform of EU rules, FI has a penchant for firms and old people, PD, especially under Renzi, was quite centrist.

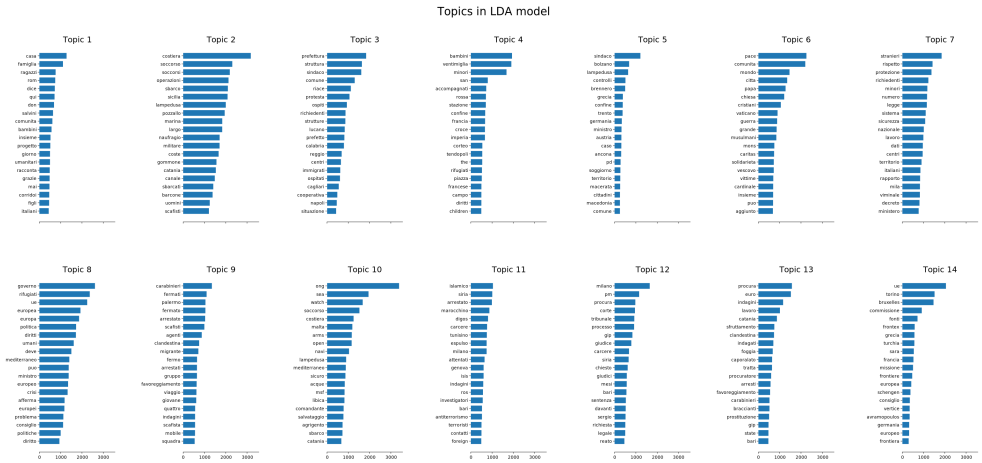


Figure C1: LDA classification of immigration news into categories

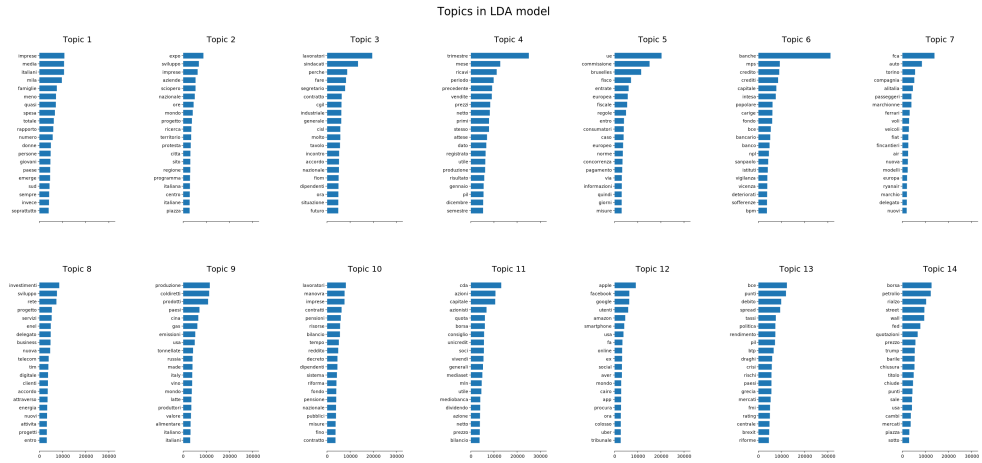


Figure C2: LDA classification of economic news into categories

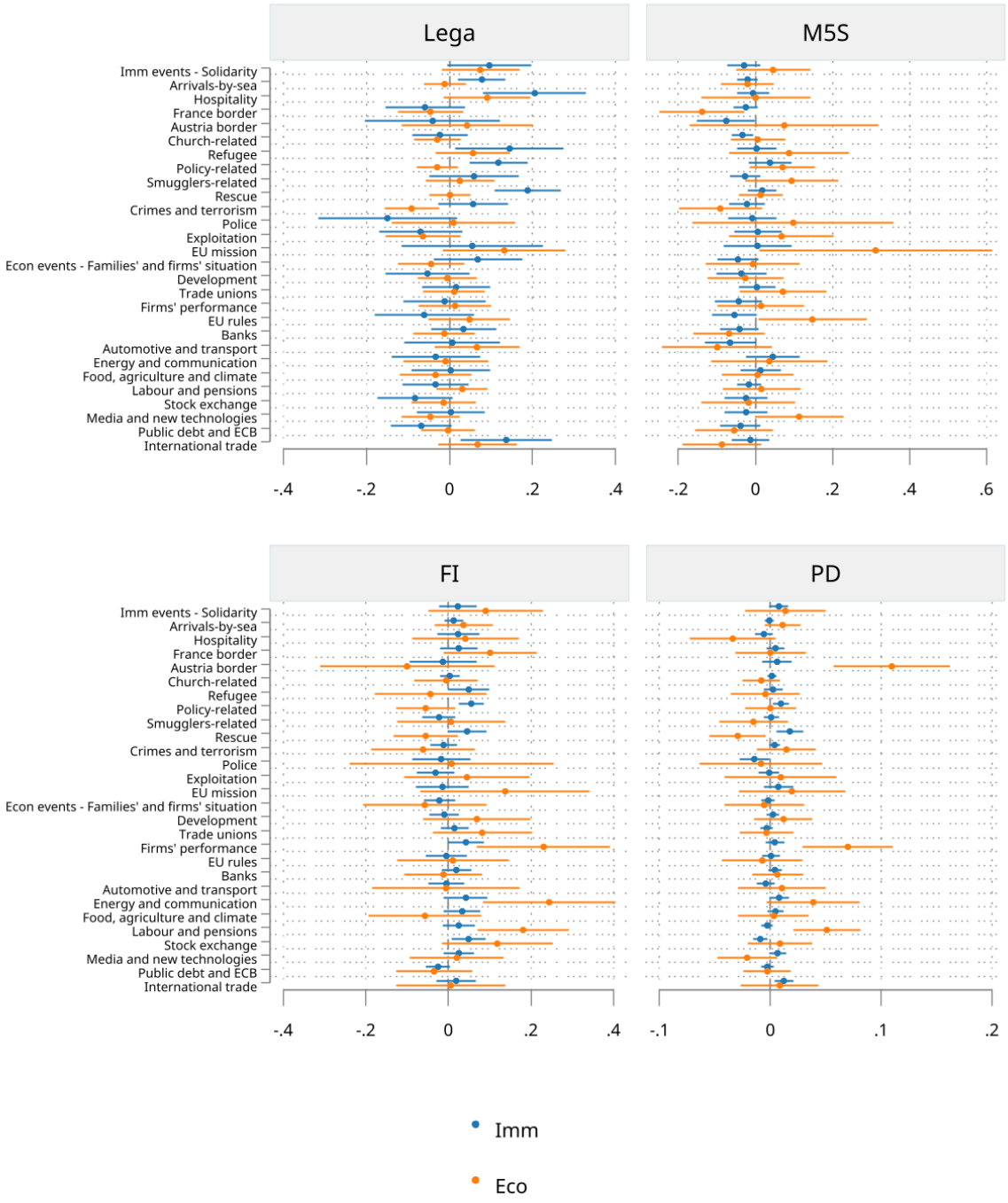
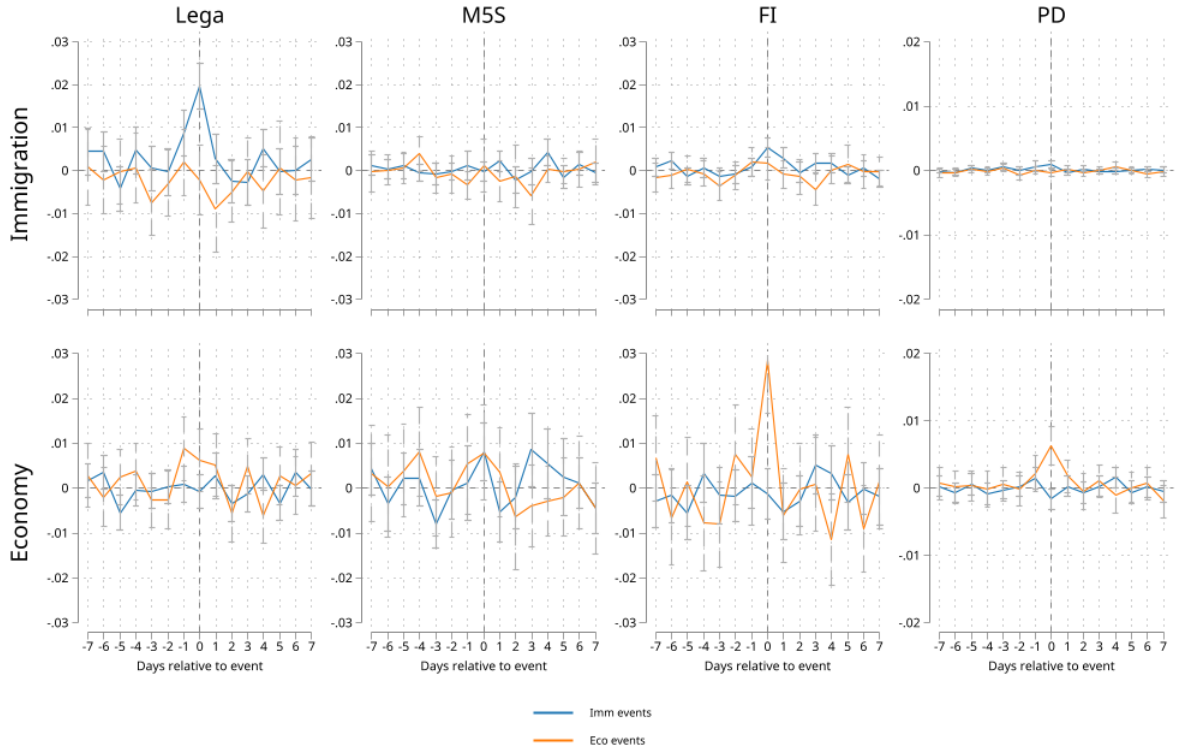


Figure C3: Impact of News by type of Event on Social Media Usage by Political Parties

D Additional results

Figure D1: Social media messages over time and events

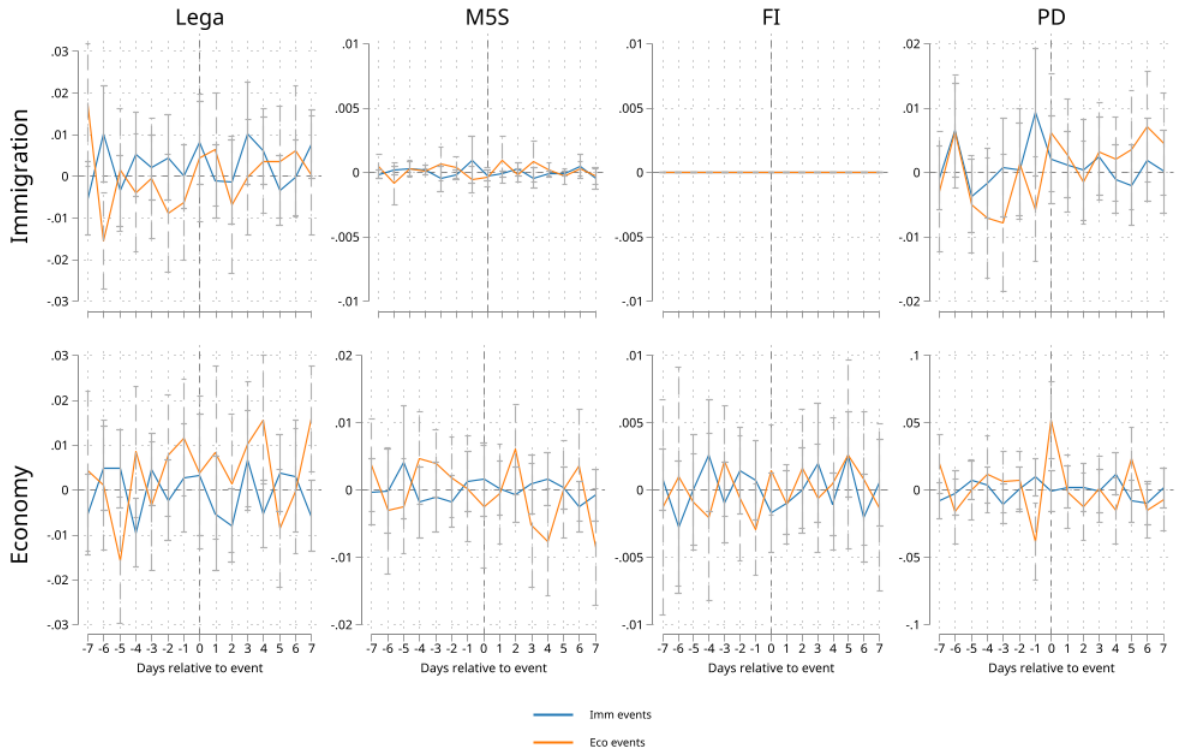


Results come from daily-level OLS regressions of events on Social Media Messages. The explanatory variables are immigration and economic events with seven lags (on the right) and leads (on the left). Errors from which the 95% confidence intervals in the graph are computed are robust with HAC covariance with bandwith equal to 6. All variables on events are standardized to be mean zero and standard deviation one across the sample. All variables on messages are standardized with respect to the total number of messages on that topic. All regressions include two lags of the total number of messages, day of week, week of year, and year fixed effects as well as daily time trends that vary by year.

Table D1: Most frequent bigrams by political party

	Lega					
	Economy			Immigration		
1	legge	fornero	836	salvini	matteo	556
2	lega	nord	652	lega	nord	436
3	flat	tax	549	ius	soli	410
4	euro	milioni	474	renzi	alfano	357
5	euro	miliardi	331	salvini	migranti	347
6	milioni	italiani	324	profughi	presunti	335
7	salvini	matteo	324	simona	bordonali	306
8	salvini	governo	316	asilo	richiedenti	224
9	salvini	portaaporta	253	immigrazione	sicurezza	203
10	renzi	governo	230	salvini	lega	201
11	posti	lavoro	224	clandestina	immigrazione	194
12	studi	settore	203	immigrati	clandestini	193
13	meno	tasse	197	immigrati	salvini	164
14	made	italy	196	massimiliano	fedriga	164
15	siri	armando	183	finti	profughi	158
	M5S					
	Economy			Immigration		
1	movimento	stelle	764	pd	sindaco	88
2	reddito	cittadinanza	695	sindaco	arrestato	58
3	maio	luigi	544	pd	brenta	54
4	euro	milioni	469	movimento	stelle	48
5	posti	lavoro	244	ballardindimettiti	brenta	43
6	italia	stelle	195	de	arrestato	41
7	euro	miliardi	184	gaetano	de	41
8	ecco	cosa	182	partitodetenuti	pd	40
9	milioni	italiani	171	partitodetenuti	arrestato	40
10	salario	minimo	145	ragione	ong	35
11	persone	milioni	130	movimento	ragione	34
12	elettorale	legge	130	stelle	chiedeteciscusa	33
13	legge	proposta	126	noallevostreguerre	guerra	32
14	euro	mila	124	migranti	cihannomentito	31
15	dopo	anni	111	ue	multa	30
	FI					
	Economy			Immigration		
1	forza	italia	1048	ius	soli	119
2	meno	tasse	512	immigrazione	sicurezza	117
3	posti	lavoro	497	immigrati	gasparripdl	96
4	reddito	cittadinanza	461	forza	italia	81
5	flat	tax	443	migranti	renatobrunetta	56
6	governo	berlusconi	324	maurizio	gasparri	54
7	renzi	governo	307	clandestina	immigrazione	41
8	legge	proposta	263	asilo	richiedenti	40
9	silvio	berlusconi	256	migranti	gruppoficamera	37
10	pressione	fiscale	218	problema	immigrazione	36
11	elettorale	legge	195	renzi	governo	34
12	debito	pubblico	188	migranti	italia	33
13	legge	bilancio	179	emergenza	immigrazione	33
14	renatobrunetta	governo	169	immigrazione	gruppoficamera	33
15	famiglie	imprese	161	italia	clandestini	31
	PD					
	Economy			Immigration		
1	partito	democratico	278	paesi	altri	18
2	renzi	matteo	252	renzi	matteo	18
3	posti	lavoro	250	guardia	costiera	13
4	legge	bilancio	207	marco	minniti	13
5	nicola	zingaretti	189	vite	umane	11
6	reddito	cittadinanza	169	donne	uomini	11
7	salvini	maio	168	consiglio	europeo	10
8	euro	miliardi	164	giusto	sbagliato	10
9	martina	maurizio	154	trattato	dublino	10
10	stelle	cinque	141	parlare	immigrazione	9
11	elettorale	campagna	138	vite	salvare	9
12	euro	milioni	115	europei	paesi	8
13	redditi	dichiarazione	102	lia	quartapelle	8
14	avanti	insieme	98	cooperazione	internazionale	8
15	giorno	ogni	97	umani	esseri	7

Figure D2: Traditional media messages over time and events



Results come from daily-level OLS regressions of events on Traditional Media Messages. The explanatory variables are immigration and economic events with seven lags (on the right) and leads (on the left). Errors from which the 95% confidence intervals in the graph are computed are robust with HAC covariance with bandwith equal to 6. All variables on events are standardized to be mean zero and standard deviation one across the sample. All variables on messages are standardized with respect to the total number of messages on that topic. All regressions include two lags of the total number of messages, day of week, week of year, and year fixed effects as well as daily time trends that vary by year.

Table D2: Impact of Important News on Social Media Usage by Political Parties

	Leftwing				*Rightwing*			
	Social media		*Traditional media*		*Social media*		*Traditional media*	
Populist parties	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	M5S on Imm	M5S on Eco	M5S on Imm	M5S on Eco	Lega on Imm	Lega on Eco	Lega on Imm	Lega on Eco
Important Eco events	-0.001	-0.004	-0.000	-0.002	-0.004*	0.001	-0.001	0.003
	(0.001)	(0.003)	(0.000)	(0.002)	(0.002)	(0.002)	(0.004)	(0.006)
Important Imm events	0.001	0.003	-0.000	-0.001	0.022***	-0.001	0.017***	0.001
	(0.001)	(0.002)	(0.000)	(0.001)	(0.003)	(0.001)	(0.005)	(0.006)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.095	0.226	0.041	0.058	0.325	0.261	0.135	0.132
Traditional parties	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	PD on Imm	PD on Eco	PD on Imm	PD on Eco	FI on Imm	FI on Eco	FI on Imm	FI on Eco
Important Eco events	-0.000**	0.003***	-0.004*	0.013	-0.000	0.005	0.000***	0.001
	(0.000)	(0.001)	(0.003)	(0.009)	(0.001)	(0.004)	(0.000)	(0.002)
Important Imm events	0.002***	0.000	0.003*	0.004	0.006***	-0.006**	0.000	-0.000
	(0.000)	(0.001)	(0.002)	(0.005)	(0.001)	(0.003)	(0.000)	(0.001)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.168	0.469	0.070	0.140	0.350	0.428	-77.274	0.049

Notes: Each column corresponds to an OLS regression. We consider here only events that are present in the ANSA evening recap of the relevant daily news. All variables on events are standardized to be mean zero and standard deviation one across the sample. All variables on messages are standardized with respect to the total number of messages. All regressions include two lags of the total number of messages, day of week, week of year, and year fixed effects as well as daily time trends that vary by year. SEs robust to arbitrary heteroskedasticity and within-panel autocorrelation and cross-panel autocorrelated disturbances that disappear after 6 lags. *, **, ***, indicate statistical significance at the 10%, 5%, and 1% level.

Table D3: Impact of News on Social Media Usage by Political Parties by Social Media

	Leftwing				*Rightwing*			
	Facebook		*Twitter*		*Facebook*		*Twitter*	
Populist parties	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	M5S on Imm	M5S on Eco	M5S on Imm	M5S on Eco	Lega on Imm	Lega on Eco	Lega on Imm	Lega on Eco
Economic events	-0.001	0.012**	0.000	0.004	-0.000	0.019***	-0.002	0.007***
	(0.001)	(0.005)	(0.004)	(0.005)	(0.006)	(0.006)	(0.003)	(0.002)
Immigration events	0.002**	0.001	0.001	0.008**	0.039***	-0.003	0.016***	0.001
	(0.001)	(0.003)	(0.002)	(0.003)	(0.005)	(0.003)	(0.002)	(0.001)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.073	0.416	0.098	0.180	0.341	0.302	0.344	0.264
Traditional parties	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	PD on Imm	PD on Eco	PD on Imm	PD on Eco	FI on Imm	FI on Eco	FI on Imm	FI on Eco
Economic events	-0.001	0.016***	-0.000**	0.002***	-0.001	0.008**	0.003	0.032***
	(0.001)	(0.004)	(0.000)	(0.001)	(0.001)	(0.003)	(0.002)	(0.007)
Immigration events	0.003***	-0.003	0.001***	0.000	0.002***	-0.001	0.008***	-0.005
	(0.001)	(0.002)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.003)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.149	0.488	0.104	0.236	0.068	0.465	0.371	0.403

Notes: Each column corresponds to an OLS regression. All variables on events are standardized to be mean zero and standard deviation one across the sample. All variables on messages are standardized with respect to the total number of messages. All regressions include two lags of the total number of messages, day of week, week of year, and year fixed effects as well as daily time trends that vary by year. SEs robust to arbitrary heteroskedasticity and within-panel autocorrelation and cross-panel autocorrelated disturbances that disappear after 6 lags.

Table D4: Impact of News on Social Media Usage by Period

	(1) Lega on imm	(2) Lega on eco	(3) M5S on imm	(4) M5S on eco	(5) FI on imm	(6) FI on eco	(7) PD on imm	(8) PD on eco
Economic events	0.005	0.012*	-0.001	-0.008	0.002	0.011*	-0.001*	0.002
	(0.007)	(0.007)	(0.003)	(0.006)	(0.002)	(0.007)	(0.001)	(0.001)
Immigration events	0.023**	-0.002	0.001	0.001	0.001	-0.005	0.003**	-0.001*
	(0.010)	(0.004)	(0.002)	(0.004)	(0.002)	(0.005)	(0.001)	(0.001)
Period								
Refugee crisis	0.045**	0.030**	0.000	-0.022	-0.013*	0.026	-0.002**	-0.017***
	(0.023)	(0.013)	(0.010)	(0.023)	(0.007)	(0.023)	(0.001)	(0.004)
Electoral campaign	0.060**	0.077***	-0.018	-0.034	0.040*	0.314***	-0.000	0.017*
	(0.030)	(0.020)	(0.013)	(0.032)	(0.021)	(0.080)	(0.002)	(0.009)
Conte I government	0.114***	0.045**	-0.002	0.007	-0.014	0.096**	0.001	-0.028***
	(0.032)	(0.020)	(0.011)	(0.027)	(0.012)	(0.046)	(0.002)	(0.008)
Period # Economic events								
Refugee crisis	-0.003	0.002	0.002	0.019***	0.002	0.014***	0.001*	0.004***
	(0.006)	(0.007)	(0.002)	(0.006)	(0.002)	(0.005)	(0.001)	(0.001)
Electoral campaign	0.009	-0.001	-0.001	0.008	0.020**	0.077**	0.000	0.007**
	(0.009)	(0.012)	(0.003)	(0.007)	(0.009)	(0.032)	(0.001)	(0.003)
Conte I government	-0.017**	-0.006	-0.001	0.012**	-0.004**	0.036***	0.001	0.012***
	(0.007)	(0.007)	(0.001)	(0.005)	(0.002)	(0.008)	(0.001)	(0.002)
Period # Immigration events								
Refugee crisis	-0.003	-0.000	-0.000	0.010*	0.008***	-0.000	-0.002*	0.002*
	(0.011)	(0.005)	(0.003)	(0.006)	(0.002)	(0.006)	(0.001)	(0.001)
Electoral campaign	-0.029	-0.011	-0.000	-0.008	0.003	0.015	0.001	0.015
	(0.023)	(0.019)	(0.007)	(0.018)	(0.019)	(0.087)	(0.002)	(0.010)
Conte I government	0.014	0.005	-0.001	-0.001	0.008**	-0.001	-0.000	0.002
	(0.011)	(0.005)	(0.002)	(0.005)	(0.003)	(0.008)	(0.001)	(0.002)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.360	0.280	0.099	0.237	0.398	0.506	0.195	0.472

Notes: Each column corresponds to an OLS regression. All variables on events are standardized to be mean zero and standard deviation one across the sample. All variables on messages are standardized with respect to the total number of messages. All regressions include two lags of the total number of messages, day of week, week of year, and year fixed effects as well as daily time trends that vary by year. SEs robust to arbitrary heteroskedasticity and within-panel autocorrelation and cross-panel autocorrelated disturbances that disappear after 6 lags. *, **, ***, indicate statistical significance at the 10%, 5%, and 1% level.

Table D5: Impact of News on Traditional Media Usage by Period

	(1) Lega on imm	(2) Lega on eco	(3) M5S on imm	(4) M5S on eco	(5) FI on imm	(6) FI on eco	(7) PD on imm	(8) PD on eco
Economic events	0.020**	0.000	-0.005	0.007	-0.000	-0.001	0.019**	0.046*
	(0.009)	(0.007)	(0.005)	(0.008)	(0.000)	(0.002)	(0.009)	(0.028)
Immigration events	-0.002	-0.002	0.000	0.006	-0.000**	-0.002	0.003	0.012
	(0.004)	(0.004)	(0.001)	(0.006)	(0.000)	(0.003)	(0.006)	(0.023)
Period								
Refugee crisis	0.024	-0.006	-0.002	-0.002	-0.000***	0.010	0.024	-0.022
	(0.027)	(0.019)	(0.003)	(0.017)	(0.000)	(0.008)	(0.019)	(0.052)
Electoral campaign	0.030	0.067	-0.004	0.013	-0.000***	0.031	0.024	0.145
	(0.039)	(0.050)	(0.005)	(0.029)	(0.000)	(0.025)	(0.027)	(0.111)
Conte I government	0.153***	-0.003	-0.003	-0.005	-0.000***	0.028	0.034	0.033
	(0.047)	(0.043)	(0.003)	(0.027)	(0.000)	(0.019)	(0.022)	(0.068)
Period # Economic events								
Refugee crisis	-0.012*	0.005	0.005	-0.011	-0.000	-0.000	-0.010	-0.003
	(0.007)	(0.006)	(0.005)	(0.007)	(0.000)	(0.002)	(0.009)	(0.026)
Electoral campaign	0.027	0.025	0.006	0.001	0.000***	-0.010	-0.012	-0.027
	(0.016)	(0.018)	(0.006)	(0.009)	(0.000)	(0.012)	(0.009)	(0.045)
Conte I government	-0.020	0.025**	0.006	-0.008	0.000***	0.004	-0.017**	-0.028
	(0.013)	(0.012)	(0.005)	(0.008)	(0.000)	(0.004)	(0.008)	(0.026)
Period # Immigration events								
Refugee crisis	0.004	-0.001	0.000	-0.001	0.000	0.002	0.008	-0.016
	(0.005)	(0.005)	(0.001)	(0.007)	(0.000)	(0.003)	(0.008)	(0.025)
Electoral campaign	-0.001	0.067	-0.002	0.001	0.000***	0.038	0.010	0.164
	(0.025)	(0.045)	(0.002)	(0.030)	(0.000)	(0.027)	(0.022)	(0.116)
Conte I government	0.046***	0.014	-0.000	-0.012*	0.000***	-0.002	-0.001	-0.008
	(0.015)	(0.017)	(0.001)	(0.007)	(0.000)	(0.004)	(0.007)	(0.024)
N	1686	1686	1686	1686	1686	1686	1686	1686
r2	0.164	0.141	0.053	0.066	-23.082	0.054	0.087	0.154

Notes: Each column corresponds to an OLS regression. All variables on events are standardized to be mean zero and standard deviation one across the sample. All variables on messages are standardized with respect to the total number of messages. All regressions include two lags of the total number of messages, day of week, week of year, and year fixed effects as well as daily time trends that vary by year. SEs robust to arbitrary heteroskedasticity and within-panel autocorrelation and cross-panel autocorrelated disturbances that disappear after 6 lags. *, **, ***, indicate statistical significance at the 10%, 5%, and 1% level.

Table D6: Impacts of Social media messages Political Parties on Voting, Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	5KM < distance < 20KM					5KM < distance < 12.5KM				
	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout
4G Coverage X Past week's econ events	0.00664**	-0.00590	-0.00157	-0.00681	0.00857	0.00632	-0.0116	-0.00885	0.0397*	0.0204
	(0.00269)	(0.00443)	(0.00234)	(0.00534)	(0.00573)	(0.0145)	(0.0213)	(0.0136)	(0.0221)	(0.0240)
4G Coverage X Past week's imm events	-0.000495	0.00332	-0.00237	0.00511	-0.00818	0.00193	-0.00429	0.0162	-0.0179	0.0167
	(0.00240)	(0.00377)	(0.00235)	(0.00437)	(0.00623)	(0.0143)	(0.0192)	(0.0137)	(0.0216)	(0.0261)
4G Coverage X Past week's imm messages by Lega	0.0452*	-0.0464	0.00890	-0.0626	0.0878	0.349**	0.133	-0.0377	-0.195	0.381
	(0.0257)	(0.0402)	(0.0256)	(0.0475)	(0.0637)	(0.150)	(0.205)	(0.127)	(0.233)	(0.280)
4G Coverage X Past week's econ messages by Lega	-0.000455	0.000638	-0.0945***	0.0883	-0.275***	0.215	-0.110	-0.188	-0.221	-0.0778
	(0.0510)	(0.0674)	(0.0363)	(0.0597)	(0.0826)	(0.274)	(0.374)	(0.229)	(0.392)	(0.478)
4G Coverage X Past week's other messages by Lega	-0.00797	0.000536	0.0130*	-0.0254*	0.0266	-0.0477	-0.00777	0.0570	0.0374	0.0517
	(0.00870)	(0.0125)	(0.00677)	(0.0133)	(0.0175)	(0.0553)	(0.0818)	(0.0370)	(0.0694)	(0.0880)
Distance interactions	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
F-Stat of sum of same-party communication across types	4.207**	2.227	0.112	0.050		3.602*	1.880	0.080	0.094	
F-Stat of all same-party communication across types	1.543	0.420	0.247	2.049	2.005**	2.213*	1.452	1.712	3.648**	0.919
N	72601	72601	72601	72601	56986	51047	51047	51047	51047	40207

Notes: This table shows the coefficients of the regression estimating Equation 3 in columns (1-4), with a minimum distance threshold of 5KM, and a maximum of 20KM in columns (1), (2), and of 10KM in columns (3) and (4). Columns (5) and (8) replicate the analysis on voters predicted to be high internet users, based on a model trained on data of the European Social Survey of 2012. All of the estimations include municipality FEs, week FEs, political communication interactions for other parties (Partito Democratico, Forza Italia, Movimento 5 Stelle). Estimations reported in the first four columns also report two-way and three-way interactions between communication and events variables with 4G coverage and distance from the 4G border. Finally, we control for respondent's socio-demographic characteristics: age, age squared, dummies for education level and occupation. The last rows in the table report (i) the F-stat relative to the null hypothesis that the sum of communication coefficients for the columns' party is equal to 0 and (ii) the F-stat relative to the null that all same-party's communication coefficients are jointly 0. In the latter test, we compute the joint F-stat on all parties when the dependent variable is turnout. Standard errors are robust to heteroskedasticity. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table D7: Impacts of Social media messages Political Parties on Voting, with Matching (2-obs units)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Whole Sample					High Predicted Internet Usage				
	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout
4G Coverage X Past week's econ events	0.00254	-0.00311	-0.00318	-0.00324	0.00336	0.00450	-0.00174	-0.00618*	-0.00519	-0.00518
	(0.00382)	(0.00456)	(0.00213)	(0.00434)	(0.00771)	(0.00543)	(0.00555)	(0.00319)	(0.00580)	(0.00834)
4G Coverage X Past week's imm events	-0.00405	0.00398	0.00262	0.00448	-0.00348	-0.00940*	0.000255	0.00642*	0.00208	-0.00655
	(0.00347)	(0.00536)	(0.00204)	(0.00435)	(0.00691)	(0.00537)	(0.00719)	(0.00327)	(0.00510)	(0.00817)
4G Coverage X Past week's imm messages by Lega	0.109	-0.134	-0.0493	-0.0676	0.308	0.303	-0.0133	-0.165*	0.140	0.540**
	(0.138)	(0.107)	(0.0553)	(0.158)	(0.188)	(0.208)	(0.154)	(0.0968)	(0.195)	(0.235)
4G Coverage X Past week's econ messages by Lega	0.187	-0.0735	-0.0371	-0.118	-0.0315	-0.0806	-0.301	0.0793	-0.221	0.203
	(0.286)	(0.180)	(0.0824)	(0.230)	(0.288)	(0.450)	(0.257)	(0.131)	(0.309)	(0.352)
4G Coverage X Past week's other messages by Lega	-0.00750	0.0157	0.00918	-0.0271	-0.0799**	0.0123	0.0109	0.0130	-0.0184	-0.120**
	(0.0270)	(0.0316)	(0.0199)	(0.0353)	(0.0400)	(0.0387)	(0.0434)	(0.0274)	(0.0476)	(0.0474)
4G Coverage X Past week's imm messages by PD	-0.109	0.444	0.547**	0.418	0.0853	-0.760	0.106	0.645	1.041	-0.527
	(0.432)	(0.590)	(0.269)	(0.575)	(0.645)	(0.665)	(0.766)	(0.452)	(0.781)	(0.740)
4G Coverage X Past week's econ messages by PD	-0.188	-0.219	0.0305	-0.0829	0.399	-0.274	-0.0895	-0.0912	0.187	0.619
	(0.226)	(0.320)	(0.176)	(0.292)	(0.478)	(0.325)	(0.439)	(0.244)	(0.436)	(0.645)
4G Coverage X Past week's other messages by PD	0.0221	0.135	0.0240	-0.122	-0.151	0.124	0.378*	-0.0379	-0.501***	-0.395*
	(0.124)	(0.138)	(0.0847)	(0.134)	(0.206)	(0.168)	(0.216)	(0.130)	(0.186)	(0.238)
4G Coverage X Past week's imm messages by M5S	0.306	-0.582	-0.216	-0.162	-0.908	0.813	-1.044	0.387	-0.661	-1.846**
	(0.360)	(0.516)	(0.224)	(0.426)	(0.606)	(0.585)	(0.742)	(0.383)	(0.615)	(0.730)
4G Coverage X Past week's econ messages by M5S	-0.00810	0.176	-0.0485	0.158	0.197	0.294	0.0456	-0.0664	-0.0972	0.200
	(0.132)	(0.161)	(0.0844)	(0.151)	(0.218)	(0.179)	(0.237)	(0.119)	(0.220)	(0.306)
4G Coverage X Past week's other messages by M5S	-0.00928	-0.0310	0.0172	0.00628	0.0230	-0.101***	0.0183	0.00872	0.0868**	0.0362
	(0.0248)	(0.0313)	(0.0158)	(0.0336)	(0.0410)	(0.0375)	(0.0460)	(0.0261)	(0.0428)	(0.0689)
4G Coverage X Past week's imm messages by FI	-0.105	0.222	-0.143	-0.108	0.00658	-0.122	0.789	-0.609*	0.00700	1.278
	(0.334)	(0.462)	(0.186)	(0.440)	(0.648)	(0.518)	(0.676)	(0.311)	(0.723)	(0.814)
4G Coverage X Past week's econ messages by FI	-0.183	0.198	-0.169*	0.136	0.113	-0.139	0.456***	-0.230*	0.0821	0.113
	(0.122)	(0.136)	(0.0925)	(0.155)	(0.182)	(0.148)	(0.174)	(0.133)	(0.210)	(0.240)
4G Coverage X Past week's other messages by FI	0.0318	-0.144***	0.0451**	-0.0374	-0.0389	0.00657	-0.290***	0.0670**	-0.0659	-0.0904
	(0.0390)	(0.0450)	(0.0202)	(0.0408)	(0.0589)	(0.0498)	(0.0563)	(0.0324)	(0.0611)	(0.0809)
F-Stat of all same-party communication across types	1.134	0.432	1.816	0.727	3.388***	1.616	1.343	2.529*	1.689	1.926**
N	102084	102084	102084	102084	80027	50770	50770	50770	50770	39978

Notes: S*, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table D8: Impacts of Social media messages Political Parties on Voting, with Matching (3-obs units)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Whole Sample					High Predicted Internet Usage				
	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout
4G Coverage X Past week's econ events	0.00321	-0.00277	-0.00278	-0.00194	0.00370	0.00564	-0.00206	-0.00519	-0.00370	-0.00540
	(0.00380)	(0.00457)	(0.00231)	(0.00413)	(0.00727)	(0.00512)	(0.00556)	(0.00325)	(0.00584)	(0.00774)
4G Coverage X Past week's imm events	-0.00434	0.00524	0.00238	0.00229	-0.00393	-0.00938*	0.00339	0.00632**	-0.00290	-0.00252
	(0.00324)	(0.00487)	(0.00213)	(0.00387)	(0.00562)	(0.00486)	(0.00639)	(0.00307)	(0.00550)	(0.00728)
4G Coverage X Past week's imm messages by Lega	0.136	-0.214*	-0.0542	0.0308	0.272	0.330*	-0.0867	-0.161*	0.284	0.502**
	(0.128)	(0.115)	(0.0535)	(0.139)	(0.174)	(0.198)	(0.160)	(0.0945)	(0.199)	(0.204)
4G Coverage X Past week's econ messages by Lega	0.132	-0.00732	0.0118	-0.152	-0.0933	-0.194	-0.180	0.161	-0.272	0.245
	(0.264)	(0.183)	(0.0798)	(0.212)	(0.306)	(0.436)	(0.251)	(0.132)	(0.321)	(0.325)
4G Coverage X Past week's other messages by Lega	-0.00886	0.0237	0.00719	-0.0343	-0.0786**	0.00158	0.0101	0.0167	-0.00553	-0.136***
	(0.0262)	(0.0280)	(0.0210)	(0.0328)	(0.0384)	(0.0360)	(0.0400)	(0.0286)	(0.0476)	(0.0444)
4G Coverage X Past week's imm messages by PD	-0.200	0.513	0.499*	0.395	-0.434	-0.856	0.302	0.767	1.004	-0.734
	(0.377)	(0.547)	(0.270)	(0.506)	(0.597)	(0.599)	(0.693)	(0.472)	(0.759)	(0.738)
4G Coverage X Past week's econ messages by PD	-0.147	-0.445	0.0208	0.211	0.457	-0.0853	-0.435	-0.186	0.489	0.705
	(0.246)	(0.310)	(0.182)	(0.280)	(0.428)	(0.308)	(0.419)	(0.238)	(0.423)	(0.624)
4G Coverage X Past week's other messages by PD	-0.0195	0.158	0.0350	-0.192	-0.208	0.118	0.424**	-0.0106	-0.575***	-0.398*
	(0.123)	(0.127)	(0.0858)	(0.126)	(0.179)	(0.154)	(0.209)	(0.126)	(0.183)	(0.221)
4G Coverage X Past week's imm messages by M5S	0.102	-0.339	-0.315	-0.162	-0.560	0.546	-0.592	0.103	-0.606	-1.612**
	(0.317)	(0.491)	(0.236)	(0.397)	(0.566)	(0.528)	(0.651)	(0.433)	(0.632)	(0.688)
4G Coverage X Past week's econ messages by M5S	0.0297	0.223	-0.0736	0.0914	0.196	0.270	0.105	-0.106	-0.223	0.0752
	(0.132)	(0.148)	(0.0871)	(0.140)	(0.182)	(0.176)	(0.225)	(0.125)	(0.223)	(0.260)
4G Coverage X Past week's other messages by M5S	-0.000994	-0.0517*	0.0227	0.0241	0.0220	-0.0902**	-0.0106	0.00984	0.119***	0.0468
	(0.0255)	(0.0287)	(0.0164)	(0.0327)	(0.0364)	(0.0360)	(0.0419)	(0.0269)	(0.0408)	(0.0579)
4G Coverage X Past week's imm messages by FI	0.182	0.141	-0.0207	-0.281	0.263	0.124	0.393	-0.394	-0.211	1.258
	(0.321)	(0.436)	(0.216)	(0.432)	(0.656)	(0.491)	(0.651)	(0.321)	(0.720)	(0.766)
4G Coverage X Past week's econ messages by FI	-0.198	0.151	-0.153	0.105	0.170	-0.115	0.418**	-0.240	0.0483	0.148
	(0.123)	(0.132)	(0.105)	(0.142)	(0.168)	(0.137)	(0.169)	(0.149)	(0.191)	(0.240)
4G Coverage X Past week's other messages by FI	0.0261	-0.138***	0.0364	-0.0235	-0.0695	-0.00165	-0.268***	0.0545*	-0.0487	-0.0829
	(0.0396)	(0.0411)	(0.0221)	(0.0392)	(0.0541)	(0.0484)	(0.0515)	(0.0320)	(0.0560)	(0.0778)
F-Stat of all same-party communication across types	1.403	0.884	1.149	0.821	1.592*	1.628	1.379	1.591	3.312**	2.360***
N	137430	137430	137430	137430	108019	68300	68300	68300	68300	53900

Notes: S*, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table D9: Impacts of Social media messages Political Parties on Voting, with Matching (4-obs units)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Whole Sample					High Predicted Internet Usage				
	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout
4G Coverage X Past week's econ events	0.00145	-0.00227	-0.00333	-0.00210	0.00136	0.00301	0.000758	-0.00577*	-0.00307	-0.00644
	(0.00426)	(0.00534)	(0.00232)	(0.00419)	(0.00692)	(0.00470)	(0.00532)	(0.00323)	(0.00586)	(0.00725)
4G Coverage X Past week's imm events	-0.00312	0.00503	0.00280	0.00172	-0.00400	-0.00750	0.00331	0.00687**	-0.00354	-0.00251
	(0.00300)	(0.00465)	(0.00232)	(0.00390)	(0.00524)	(0.00461)	(0.00593)	(0.00302)	(0.00512)	(0.00701)
4G Coverage X Past week's imm messages by Lega	0.100	-0.210**	-0.0585	0.0242	0.270	0.270	-0.0586	-0.173*	0.300	0.495**
	(0.125)	(0.106)	(0.0536)	(0.138)	(0.164)	(0.183)	(0.154)	(0.0894)	(0.195)	(0.196)
4G Coverage X Past week's econ messages by Lega	0.162	-0.0105	0.00126	-0.153	-0.0818	-0.131	-0.188	0.201	-0.291	0.315
	(0.267)	(0.183)	(0.0831)	(0.204)	(0.277)	(0.405)	(0.238)	(0.136)	(0.314)	(0.312)
4G Coverage X Past week's other messages by Lega	-0.0112	0.0250	0.00979	-0.0330	-0.0750**	0.00243	0.0117	0.0163	-0.0160	-0.146***
	(0.0256)	(0.0273)	(0.0214)	(0.0316)	(0.0372)	(0.0349)	(0.0369)	(0.0276)	(0.0469)	(0.0407)
4G Coverage X Past week's imm messages by PD	-0.118	0.392	0.356	0.401	-0.476	-0.607	0.249	0.631	1.286*	-0.786
	(0.367)	(0.553)	(0.271)	(0.496)	(0.619)	(0.559)	(0.644)	(0.459)	(0.775)	(0.713)
4G Coverage X Past week's econ messages by PD	-0.148	-0.425	0.0748	0.180	0.460	-0.194	-0.355	-0.158	0.386	0.769
	(0.234)	(0.303)	(0.184)	(0.281)	(0.406)	(0.317)	(0.383)	(0.245)	(0.418)	(0.604)
4G Coverage X Past week's other messages by PD	0.0122	0.150	0.00257	-0.185	-0.178	0.166	0.325*	-0.0580	-0.475**	-0.371*
	(0.121)	(0.129)	(0.0881)	(0.129)	(0.177)	(0.150)	(0.194)	(0.125)	(0.187)	(0.212)
4G Coverage X Past week's imm messages by M5S	0.0560	-0.295	-0.202	-0.377	-0.617	0.511	-0.781	0.142	-0.777	-1.579**
	(0.308)	(0.494)	(0.225)	(0.384)	(0.580)	(0.485)	(0.621)	(0.386)	(0.617)	(0.631)
4G Coverage X Past week's econ messages by M5S	0.0547	0.288*	-0.0630	0.0553	0.152	0.290*	0.211	-0.0812	-0.316	-0.0349
	(0.133)	(0.146)	(0.0812)	(0.140)	(0.187)	(0.165)	(0.220)	(0.114)	(0.218)	(0.259)
4G Coverage X Past week's other messages by M5S	-0.0104	-0.0490*	0.0216	0.0273	0.0316	-0.103***	-0.0126	0.00729	0.128***	0.0588
	(0.0246)	(0.0282)	(0.0158)	(0.0329)	(0.0348)	(0.0346)	(0.0401)	(0.0266)	(0.0426)	(0.0526)
4G Coverage X Past week's imm messages by FI	0.0951	0.190	0.0178	-0.0808	0.327	-0.0492	0.585	-0.294	-0.341	1.173
	(0.304)	(0.417)	(0.193)	(0.420)	(0.617)	(0.472)	(0.631)	(0.312)	(0.723)	(0.714)
4G Coverage X Past week's econ messages by FI	-0.214*	0.122	-0.177	0.193	0.175	-0.127	0.356**	-0.271*	0.194	0.156
	(0.118)	(0.133)	(0.108)	(0.132)	(0.158)	(0.137)	(0.163)	(0.144)	(0.188)	(0.218)
4G Coverage X Past week's other messages by FI	0.0282	-0.128***	0.0406*	-0.0484	-0.0731	0.00762	-0.248***	0.0557*	-0.0845	-0.0882
	(0.0390)	(0.0417)	(0.0219)	(0.0374)	(0.0521)	(0.0474)	(0.0522)	(0.0310)	(0.0545)	(0.0741)
F-Stat of all same-party communication across types	1.197	0.773	1.434	0.735	3.758***	1.371	0.944	1.730	3.056**	3.429***
N	170059	170059	170059	170059	133608	84740	84740	84740	84740	66929

Notes: S*, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table D10: Impacts of Social media messages Political Parties on Voting, with Matching (3-obs units)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Whole Sample					High Predicted Internet Usage				
	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout
4G Coverage X Past week's econ events	0.00398	-0.00117	-0.00168	-0.0000332	0.00297	-0.000181	-0.00264	-0.00105	0.0000296	-0.00676
	(0.00301)	(0.00365)	(0.00208)	(0.00374)	(0.00530)	(0.00463)	(0.00533)	(0.00332)	(0.00531)	(0.00656)
4G Coverage X Past week's messages on imm	0.0837	-0.0742	0.0405	-0.0267	-0.0533	0.0832	-0.0509	0.0443	0.0732	-0.00180
	(0.0524)	(0.0460)	(0.0295)	(0.0578)	(0.0718)	(0.0903)	(0.0654)	(0.0445)	(0.0768)	(0.0907)
4G Coverage X Past week's messages on econ	-0.0841	0.127**	-0.0509	0.0493	0.0707	-0.0636	0.134*	-0.0475	-0.0714	0.0466
	(0.0608)	(0.0536)	(0.0352)	(0.0672)	(0.0806)	(0.102)	(0.0767)	(0.0523)	(0.0906)	(0.103)
4G Coverage X Past week's messages on other topic	0.0104	-0.0510***	0.0122	-0.0191	-0.0309	-0.00446	-0.0647***	0.00333	0.00196	-0.0379
	(0.0137)	(0.0160)	(0.0100)	(0.0188)	(0.0213)	(0.0170)	(0.0245)	(0.0151)	(0.0270)	(0.0293)
4G Coverage X Past week's imm events	-0.00291	0.00545	0.000289	0.00436	0.00326	-0.00475	0.00615	0.00221	0.00338	0.00750
	(0.00250)	(0.00395)	(0.00168)	(0.00306)	(0.00460)	(0.00389)	(0.00528)	(0.00258)	(0.00454)	(0.00611)
N	137430	137430	137430	137430	108019	68300	68300	68300	68300	53900

Notes: S*, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table D11: Impacts of Social media messages Political Parties on Voting, with Matching (3-obs units)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Whole Sample					High Predicted Internet Usage				
	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout	Lega Poll	PD Poll	FI Poll	M5S Poll	Turnout
4G Coverage X Past week's messages by Lega	0.0206	-0.00487	0.00194	-0.0347	-0.0425	0.0108	-0.0215	0.0132	0.0101	-0.0285
	(0.0278)	(0.0205)	(0.0150)	(0.0225)	(0.0322)	(0.0400)	(0.0297)	(0.0211)	(0.0353)	(0.0402)
4G Coverage X Past week's messages by PD	-0.0390	0.169***	-0.0249	0.0468	0.0421	0.0634	0.265***	-0.0244	-0.0519	0.0102
	(0.0381)	(0.0401)	(0.0270)	(0.0467)	(0.0561)	(0.0483)	(0.0621)	(0.0403)	(0.0671)	(0.0729)
4G Coverage X Past week's messages by M5S	0.00861	-0.0550**	0.00280	0.0314	0.0318	-0.0578*	-0.0338	0.00337	0.114***	0.0466
	(0.0171)	(0.0228)	(0.0124)	(0.0255)	(0.0298)	(0.0311)	(0.0306)	(0.0204)	(0.0312)	(0.0439)
4G Coverage X Past week's messages by FI	0.0199	-0.103***	0.0264*	-0.0419	-0.0572	-0.000349	-0.179***	0.0128	-0.0779*	-0.0329
	(0.0301)	(0.0293)	(0.0153)	(0.0281)	(0.0427)	(0.0388)	(0.0396)	(0.0226)	(0.0441)	(0.0543)
4G Coverage X Past week's imm ratio by Lega	0.0648	-0.141	-0.0582	0.0631	0.261	0.304	0.0399	-0.196**	0.247	0.371
	(0.148)	(0.115)	(0.0509)	(0.140)	(0.196)	(0.243)	(0.152)	(0.0894)	(0.201)	(0.238)
4G Coverage X Past week's imm ratio by PD	0.105	0.531**	-0.0606	0.162	-0.156	0.00811	0.314	0.254	0.220	-0.281
	(0.198)	(0.267)	(0.148)	(0.257)	(0.355)	(0.293)	(0.353)	(0.207)	(0.377)	(0.536)
4G Coverage X Past week's imm ratio by M5S	-0.0132	-0.255*	0.0259	-0.0502	-0.143	-0.258	-0.125	0.0691	0.311	-0.0181
	(0.133)	(0.145)	(0.0900)	(0.143)	(0.190)	(0.209)	(0.224)	(0.132)	(0.221)	(0.275)
4G Coverage X Past week's imm ratio by FI	0.211	-0.192	0.158	-0.191	-0.220	0.132	-0.437**	0.179	-0.219	-0.147
	(0.134)	(0.146)	(0.0994)	(0.136)	(0.184)	(0.166)	(0.195)	(0.143)	(0.192)	(0.252)
N	137430	137430	137430	137430	108019	68300	68300	68300	68300	53900

Notes: S*, **, *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table D12: Calibrated parameters

Definition	Parameter	Estimate
Fraction of voters always voting B	θ_B	0.137
Cost of communicaiton, party A	c_A	0.000028
Cost of communicaiton, party B	c_B	0.000058
Logistic for salience, intercept	α_0	0.611
Logistic for salience, impact of imm. comm.	α_1	2.934
Logistic for salience, impact of econ. comm.	α_2	-6.908
Logistic for salience, impact of imm. events (type 1)	$\alpha_{1,m}$	0
Logistic for salience, impact of imm. events (type 2)	$\alpha_{2,m}$	0
Logistic for salience, impact of econ. events (type 1)	$\alpha_{1,n}$	-0.237
Logistic for salience, impact of econ. events (type 2)	$\alpha_{2,n}$	-0.858
Advantage of party A on imm., impact of imm. events (type 1)	$\eta_{1,m}$	0.597
Advantage of party A on imm., impact of imm. events (type 2)	$\eta_{2,m}$	0.636
Advantage of party A on econ., impact of imm. events (type 1)	$\eta_{1,n}$	0.629
Advantage of party A on econ., impact of imm. events (type 2)	$\eta_{2,n}$	0.417
Advantage of party A on imm., base coefficient	η_0^m	0.336
Advantage of party A on econ., base coefficient	η_0^n	0.153
Standard deviation of searches	σ_a	0.446
Autocorrelation of searches	ρ_a	0.615
Advantage of party A on imm., standard deviation	σ^m	0.406
Advantage of party A on econ., standard deviation	σ^n	0.137